

PAKISTAN GEMS AND JEWELRY INSTITUTE - ELP

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FINAL
REPORT

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Executive Summary

This report presents the outcomes of the primary and secondary research we, students of final year at IBA (BBA), conducted for the revival of the Pakistan Gems and Jewelry Institute located in Karachi. The institute comes under the ownership of Trade Development Authority of Pakistan (TDAP) and is non-operational since the past 15 years and now realizing the potential of the gems and jewelry industry, thoughts are being given to reopen the institute.

In the secondary research, we mainly made use of the credible sources on the internet to gather information about the current market structure and the demand and supply mechanisms of the market. We also made use of the IBA research database to gather factual and quantitative data for getting upon good conclusions. Gathering the data about the machines to be installed in the institute was one the most important part and for that we used sources as Alibaba.com (Chinese online delivery giant) to get the current market prices of the demanded machineries.

The primary research was conducted to get the views of the people directly working in the market. For that, we stepped into the manufacturing and the retailing side of the businesses and interacted with the business community. Surveys and in-depth interviews were the main sources of data gathering techniques we used and we later analyzed the data gather in both, quantitative and qualitative manner.

The report elaborates on three major options we present to the TDAP for operating this institute:

- 1) Running it as an incubation facility
- 2) Public - Private Partnership
- 3) Running it as a training Institute.

All the three points given above are mentioned in great detail in the report and their pros and cons are mentioned with them accordingly. Additionally, the costing of the project is also attached with the report which gives an idea of the future expansion plans and the costs associated with it. The costing also projects the self-sustainability revenues generated by the project through different investments made.

SECONDARY RESEARCH

Introduction:

There is historical evidence that gems and precious stones have been cherished and traded for thousands of years in the sub-continent and especially in the areas comprising of modern day Pakistan. We can see this from looking into the relics, gems and jewelry that belonged with the Indus valley and Gandhara civilizations. Over the course of time, jewels were developed and flourished in the Indian sub-continent.

The South Asian sub-continent is known throughout the world as a treasure chest of gems and jewelry. Pakistan has emerged as the fifth largest source for gemstones on the world map. There is an abundance of resources of precious and semi-precious gemstones in the Northern Areas, Azad Jammu and Kashmir (AJK) and Khyber Pakhtunkhwa (KPK), with significant potential in Balochistan. Pakistan is famous for the emerald of Mingora (Swat), the pink and golden topaz of Katlang (Mardan) and the aquamarine of Chitral and Neelum Valley.

The gemstones that are mined in Pakistan are either selected for use in jewelry, for which they must be cut and polished; or for marketing as "specimens" for collectors and museums, for which they are left with the source rock. **Exhibit 1** shows the approximate distribution of Gems and Jewelry. The large domestic market is driven by demand for 22kt traditional jewelry, which is perceived as a form of investment. However, international fashion trends now wield a strong influence on demand by the domestic elite. Each of the major cities of Pakistan has a **"Sarafa Bazaar"** (jewelry market), consisting of hundreds of small showrooms, bullion dealers and casting shops. Presently, the metropolitan cities of Lahore and Karachi are the major hubs of gold, silver and alloy-based jewelry manufacturing. Other important centers are Lahore, Rawalpindi, Hyderabad, Faisalabad and Peshawar especially for Gems.

The industry itself is highly fragmented, with very few players having complete in-house production facilities. Most manufacturing is outsourced to small vendors, with family workshops employing two to four people. Women entrepreneurs and producers are currently invisible in the gems and jewelry sub-sector.

Khyber Pakhtunkhwa

“Hindukush covers the area to north and north-west, Karakoram to the north and north-east, and Himalayas to the east. According to Bureau of Statistics of Khyber Pakhtunkhwa, 2568 tonnes of baryte and 85 tonnes of corundum were produced in 2005-2006 and 1416 tonnes of quartz was produced in 2006-2007 in the province. According to one source, Swat has reserves of 70 million carats of emerald, Mardan has reserves of 9 million carats of pink topaz.”

Federally Administered Tribal Areas

“The Federally Administered Tribal Areas are strategically located between the Afghanistan and Khyber Pakhtunkhwa province of Pakistan. The region is mostly dry and barren with hilly northern Bajaur and Mohmand agencies. In the central agencies of Khyber, Kurram, and Orakzai, the Safed Koh range is located while in the two southern agencies of Waziristan, the Sulaiman range along with Waziristan hills are located. A large variety of minerals and gemstones are found in these mountains: emerald and tourmaline are found in the north, and garnet and quartz are found in the Bajaur and South Waziristan agencies.”

Gilgit Baltistan

This region has three mountain ranges; Himalayas, Hindukush, and Karakoram contain many minerals and gems including emerald, ruby, sapphire, aquamarine, moonstone, and amethyst. A number of other minerals are also found in the region such as peridot, tourmaline, topaz, garnet, red spinel, pargasite, diopside, sphene, apatite, azurite, rose quartz, and agate. In Swat, pale green to green coloured emeralds can be found in talc-carbonate schist. In Hunza, well-formed pink to red crystals of ruby are found, while in Neelum valley high quality rubies also occur.

Balochistan

It is the largest province of Pakistan by area and is covered by rough terrain and rugged mountain ranges. Major mountain ranges of the province include Makran, Sulaiman, Toba Kakar, and Kirthar. The main gemstones that are traded in the region include emerald, apatite, sapphire, agate, tourmaline, ruby, topaz, turquoise, lapis lazuli, quartz, garnet, and peridot.

Potential:

Despite its abundant reserves of precious and semi-precious gemstones and rich history of jewelry manufacturing, Pakistan has been unable to develop an internationally competitive gems

and jewelry industry. Capitalizing on its vast natural resources, low labor costs, and skilled craftsmen and growing national and international demand, Pakistan has the potential to position itself as a regional hub for precious stone cutting and jewelry manufacturing. Developing this potential will have a significant impact on Pakistan's economy in terms of increase in employment and entrepreneurship, income generation, export revenues, and poverty alleviation. Consisting of mainly small and medium entities, growth of this sector will also have positive externalities for social indicators such as health and education.

Due to lack of adequate processing facilities and skills, limited understanding of gemology and lack of standardization and certification, approximately 75 percent of Pakistan's exports consist of un-worked stones, representing a significant loss in value added. Gemstone cutting and polishing is done on a very limited scale and that too with outdated and underdeveloped techniques. There are only two commercial certification labs in the country, which are not internationally recognized. Lack of recognized lab infrastructure often leads to underselling of precious stones as well as lack of buyer confidence. Commerce is mainly conducted on a person to person basis with no established pricing structures. This imposes a significant transaction cost on the buyers and sellers.

Despite the current constraints, the gems and jewelry sector has tremendous potential to contribute to the national economy and develop into a competitive, high value added sector. Expanding national and international markets, Pakistan's vast natural resources and relatively low labor costs offer enormous opportunities for growth. The domestic market is currently estimated at a substantial USD 1.2 billion¹⁰. On the other hand, a growing and evolving domestic market can offer tremendous opportunities for the industry to diversify and expand sales if it is able to upgrade its production and marketing capabilities.

A growing international market for gems and jewelry indicates that there is significant potential for Pakistan's exports. Year on year growth for gold jewelry in the first three quarters of 2005 rose by 12 percent in tons and 20 percent in USD terms. Growth in global gold jewelry consumption in USD terms and in tonnage is given in figure 2.1. In value terms, global gold jewelry consumption grew by 18 percent from 2003 to 2004 and by 20 percent over January – October 2004 and 2005. Global gold jewelry consumption in tons rose by 6 percent over 2003-

2004 and by 12 percent over January – October 2004 and 2005. India is the world's largest gold jewelry market by volume accounting for around 520 tons of consumption demand in 2004.

In terms of consumption demand, the USA is the world's second largest gold jewelry market by volume (around 350 tons) and the largest by retail value (around \$16 billion)¹¹. Demand for natural gemstones is also growing, partly due to the emergence of mass merchandisers entering the fine jewelry market. Pakistan has a potential yield of 800,000 carats of ruby, 875,000 carats of emerald and five million carats of peridot, which remain unutilized.

Jewelry Making Process:

Jewelry is created using many methods but the most common method is by the 'lost wax' casting method. It was developed in ancient times and it is still used to this day, although modern tools are now used.

Step 1:

The reason that is called lost wax is, because the object that is to be created is first sculpted out of wax to the exact specifications that the finished product is going to look like.

Step 2:

After the wax sculpture is completed it is then encased in a silica encasement or in the case of jewelry it is then encased in plaster.

Step 3:

Then after the plaster encasement has hardened and dried it is put into an oven or kiln. This will cause the wax to melt and burn out leaving its hollow image inside of the plaster encasement.

Step 4:

Then the plaster encasement with the hollow shape of the piece of jewelry is placed into a casting centrifuge which is a device that will use centrifugal force to drive molten gold down through an opening in the chunk of plaster where it will fill the hollow spot left by the wax melting out.

Step 5:

After it has cooled the plaster is broken open to reveal a piece of gold jewelry where the wax sculpture once was.

This method has been summarized in **Exhibit 2**. It is the same technique that is used to create larger bronze sculptures, only with bronze it is done on a larger scale and the centrifugal caster isn't used. The lost wax casting method was what was used to bring civilization out of the Iron Age and into the Bronze Age. In ancient times primitive metal smiths would use bees wax to shape objects such as axes and arrow heads. These objects were then encased in clay and placed in the bottom of fire pits to burn off.

Gem Making Process:

The process of cutting and polishing gems is called **gem-cutting** or **lapidary**, while a person who cuts and polishes gems is called a **gem-cutter** or a **lapidary** (sometimes lapidarist). Gemstone material that has not been extensively cut and polished is referred to generally as **rough**. Rough material that has been lightly hammered to knock off brittle, fractured material is said to have been **cobbed**. All gems are cut and polished by progressive abrasion using finer and finer grits of harder substances.

Diamond, the hardest naturally occurring substance, has a Mohs hardness of 10 and is used as an abrasive to cut and polish a wide variety of materials, including diamond itself. Silicon carbide, a manmade compound of silicon and carbon with a Mohs hardness of 9.5, is also widely used for cutting softer gemstones. Other compounds, such as cerium oxide, tin oxide, chromium oxide, and aluminum oxide, are frequently used in polishing gemstones. In small operations, gem stone ores are broken, crushed, and concentrated by hand picking, washing, screening, or jigging. In larger operations, mechanized processes are employed. For instance, diamond processing involves standard gravity methods, grease belts, electrostatic separation, skin-flotation, magnetic separation, separation by x-ray luminescence, and separation by optical sorting.

Several common techniques are used in lapidary work:

- Sawing
- Grinding
- Sanding
- Lapping
- Polishing

In most gem sawing, a thin circular blade usually composed of steel, copper, or a phosphor bronze alloy impregnated along the outer edge with diamond grit and rotating at several thousand surface feet per minute literally scratches its way through a gemstone. A liquid such as oil or water is used to wash away cutting debris and keep the stone and the saw blade from overheating, which could cause damage to both the stone and the sawblade. Several sizes of circular rock saws are frequently used by most gem cutters:

Sawing:

- A **slab saw**, typically 16 to 24 inches in diameter, is used to cut stones of several inches thickness into relatively thin slabs (often 1/8 to 3/8 inch thick).
- A **trim saw**, typically 6 to 10 inches in diameter, is used to cut smaller stones into thin slabs or to cut small sections out of slabs.
- A **faceter's trim saw**, typically 4 inches in diameter, is used with a very thin blade, to saw small pieces of expensive rough.

There are also jigsaws that employ either a reciprocating wire or a continuous thin metal band. These are useful for cutting curved lines that are impossible with circular saws. They are also useful in minimizing waste on extremely valuable rough material

Grinding

This is usually done with silicon carbide wheels or diamond-impregnated wheels, is used to shape gemstones to a desired rough form, called a **preform**. As with sawing, a coolant/lubricant (water or oil) is used to remove debris and prevent overheating. Very coarse diamond or silicon carbide, such as 60 grit, or mesh, (400 micron particles) or 100 grit (150 micron particles) is used for rapid removal of stone, and finer abrasive (600 grit - 30 micron, or 1200 grit - 15 micron) is used for final shaping and sanding.

Sanding

This is similar to grinding but uses finer abrasives. Its purpose is to remove deep scratches left by coarser abrasives during grinding. Since it removes material less rapidly, it also allows more delicate control over final shaping of the stone prior to polishing. For stones with rounded

surfaces, a flexible surface such as a belt sander is often used to avoid creating flat areas and promote smooth curves.

Lapping

This is very similar to grinding and sanding, except that it is performed on one side of a rotating or vibrating flat disk known as a **lap**, and it is used especially to create flat surfaces on a stone (as in faceting). Laps are often made of cast iron, steel, or a copper-bronze alloy, but other materials can also be used.

Polishing

After a gemstone is sawed and ground to the desired shape and sanded to remove rough marks left by coarser grits, it is usually **polished** to a mirror-like finish to aid light reflection from the surface of the stone (or refraction through the stone, in the case of transparent materials). Very fine grades of diamond (50,000 to 100,000 mesh) can be used to polish a wide variety of materials, but other polishing agents work well in many instances. Usually, these polishing agents are metal oxides such as aluminum oxide (alumina), cerium oxide, tin oxide, chromium oxide, ferric oxide (jeweler's rouge), or silicon dioxide (tripoli).

Different stones are often very inconsistent in their ease of polishing, particularly in the case of faceted stones, so gem-cutters are often very inventive in trying new combinations of polishing agents and polishing surfaces -- often tin, tin-lead, lead, leather, felt, pella, wood, or lucite laps for flat surfaces such as facets. Rounded surfaces, such as on cabochons, are often polished on felt, leather, cork, cloth, or wood. Polishing removes small quantities of stone and can be used, especially when faceting small stones, to do ultrafine shaping of the stone. **Exhibit 3** summarizes the entire Gem cutting process.

Gems and Jewelry Machines and Tools:

These include various simple as well as complicated machines for processing, shaping of jewelry and gems like:

Silicon Carbide Sanding belt, Flame photometer, Roma micro spectrometer, Analytical balance, Conductometer, Gravity meter, Shaking table, Distillation unit, Fluid inclusion stage, Metal

detector, XRF, High temperature furnace, Ultrasonic water bath, Gemological microscope, Bausch and lamp loup, Belt sander, Bead mills, Centrifuge, Calcite dichroscope, Diamond tester, Gems pro refractometer, Grinding kit, Heart and arrows scope, Melting point apparatus, Hannerman gem filter set, Muffle furnace, OPL spectroscope, Polishing heads face, Polishing medis, Diamond colorimeter, Polariscope, Gem carat scale, Slab saw, Trim saw, Calibrating machine, Ultrasonic drilling machine, Gem carving machine, Barrel finishing machine, Vibrasonic tumbler, Dual grinder, Expendable drum combination, Sphere making machine, Automatic cabochon machine, Lath machine, Fordom flexible shaft machine, Lath machine, Sand blaster, Diamond discs, Coper polishing laps, Tin polishing laps, Steel polishing laps, Spacer, Vinyl polishing lap, Carving tools, Sponge rubber laps, True circle diamond disc, Diamond hallow drills, Leather polishing belt, Solder flux, Pressure Casting Machine, Vacuum Casting Induction, Portable gold melting machine, Graphite Crucible, LASER Welding and Marking.

Expansion and Export Potential for Gem and Jewelry Industry in Pakistan

Pakistan sits on huge natural resources of gems and semi-precious metals. Yet however, Pakistan has so far been unable to completely and successfully capture this market. The reason behind this failure is accredited to being too focused attention of Pakistani Industry. Mastering only in craftsmanship, our industry nearly suffers in all of the processes commonly affiliated with this industry. Such processes generally encompass Mining, Processing, Designing, Value-addition processes etc. After such processes, the pieces are generally marketed and then usually exported to promising markets.

Pakistani Industry uses '**Blasting Process**' while mining for such resources, which diminishes severely the Quantitative and Qualitative value of the product. Modern and careful mining methods have unfortunately not been introduced domestically that undermines our market, and industry. Further use of obsolete technology to handle such delicate pieces of jewelry damages much of the product, with decreasing quantity as well. Since Pakistan lacks in designing capabilities as well, which usually attract investors and customers, Pakistani Industry hasn't been able to fully reach it's right potential.

Besides all these negative aspects, we must recognize the opportunities present in our country too. Pakistan, being exceptionally good in craftsmanship and holding huge deposits of precious natural Gems, shows the huge potential our country and this industry holds. The exponential growth in our exports since 2000 is a testimony to our capabilities. Following measures could very well just boost our industry's exports and production:

- Expertise of mining
- Skilled labor
- Proper use of modern technology
- Education in designing (like motioned by GJTMC)

After use of these measures, our Gems and Jewelry Industry would expand to new heights to capture world market, and export them to capture the taste of people all across from the globe. Yet however, after this proper production and value-addition through astonishing craftsmanship and designing we would need a proper channel for Marketing and Branding that would put up our work into the eyes of foreign customers. PGJDC has already committed itself to this goal, and has been working on exhibitions and such marketing activities that may benefit our goals.

Other similar opportunities present in Pakistan that really attract expansion, investments, and ability to export are:

- Gemstone deposit
- Skilled human resources and craftsmanship
- Dedicated company (PGJDC) facilitating the industry across the entire value chain
- Commitment to economy pursuing open trade and investment policies
- Investment friendly Government policies
- Large (and growing) domestic market
- Well-established infrastructure and regulatory framework for Gem Industry
- Equitable fiscal regime
- Strategic location of Pakistan easing trade
- Liberalization policies

Skills and Expertise of Pakistani labor in Gem and Jewelry Industry

The skills and expertise are quite popular in all of globe in terms of craftsmanship. The precisions with which pieces of jewelry are put together to form something seem to have been inherited since the times of Mogul Empire. But still, it seems our labor employed in Gems and Jewelry Industry hasn't really incorporated themselves with the competition and technological advancements of modern times.

Not finding this Industry as prospective, many family-inherited skilled people have left the tradition, which has faded these jewelry-making skills of Pakistani people. Presently, only in certain places of the country do these artisans exist. These people have abandoned these skills for other Industries, further closing this industry in Pakistan to the brink of extinction. For proper master training, plans are underlined to bring masters of such practices from India, to revive these skills here, and prevail this occupation in Pakistan.

Further, Gems and Jewelry Training and Manufacturing Centre has been working on its journey to upgrade and augment the human resource capital in the gems and jewelry industry in the country. GJTMC has devised a four-tier strategy that holds to support its objectives of existence of human resource development, projection of professional skills, projection of technological potential and financial viability through commercial services. This center also trains students to revive this skill and work on this industry to bring it as a potentially important for the economy of Pakistan.

Revenue generation opportunities and Formalization of Gems and Jewelry sector

Gems and Jewelry Industry in Pakistan is an emerging sector with immense potential for business development. The industry is moving on fast track to gain an important position in the world market. The country's Gems and Jewelry have shown phenomenal growth in the last few years. The industry has a promising future outlook due to vast gemstones deposits, skilled human resource and craftsmanship, investment friendly policies, large and fast growing domestic markets, strategic location and liberalization and business policies. Having been blessed with a

variety of gemstones, and being one of the top ten consumers of gold in the world the industry holds great potential to attract investment in the sector. Presence of major hubs of gems and jewelry in every major cities of the country has been augmenting the startup of business activities and helping new entrepreneurs. The robust and steadily increasing domestic demand with rise in population has been creating opportunities for the new comers to invest in Gems and Jewelry business. Transformation of old infrastructure into technology driven infrastructure coupled with availability of skilled manpower is making the gems and jewelry industry ready to compete internal and external markets.

Computer Aided Jewelry Design (CAD)

Although the Computer Aided Jewelry Design (CAD) Houses can be setup anywhere, as no raw material is required, but the most ideal location for setting up a CAD house are the Jewelry hubs of the Country, such as, Karachi, Lahore, Islamabad/Rawalpindi, Hyderabad, Sukkur, Multan, Sargodha and Faisalabad. The Jewelry markets of these areas will offer maximum consumption of the designs produced through the CAD Houses.[CITATION PGJ14 \l 1033].

Capacity: *1200 designs per year*

Cost of each design: On an average it would be **Rs. 2,500**. However, the cost of design depends largely on type of jewelry and its complexity.

Production capacity: In the first year of operation, the unit would run on **60%** of the total production capacity.

Cost: The total cost of the project is **Rs. 1.440 Million**.

Capital cost: The capital cost would be around **Rs. 1.098 Million**

Operating expenses: The operating expenses would be around **Rs. 0.342 Million**.

Internal rate of Return: It is a highly feasible business with a **13.51%** Internal Rate of Return.

Payback Period: A 3-year payback period.

Along with that Pakistan lacks Jewelry Polishing Unit that can earn an **average annual return of 23.29% with a payback of 2 years** .The total **cost of the project is Rs. 0.300 Million, with capital cost of Rs. 0.156 Million and yearly operating expenses of Rs. 0.144 Million**.[CITATION PGJ13 \l 1033].

Jewelry manufacturing is a fragmented cottage industry with a rich tradition of craftsmanship; however limited design capabilities, lack of modern manufacturing technology and techniques, and poor international and branding have prevented the industry from realizing its full potential. There are limited vocational training opportunities, no investment in research and product development and weak linkages between training institutes and the industry. Furthermore, due to lack of assaying and hallmarking, there is no institutional mechanism for providing product guarantee in the local and international market. There are more than thirty major cities and nearly three hundred smaller cities/mandi towns where jewelry manufacturing and trading clusters cater to domestic demand. In addition, there are at least 45,000 villages where jewelers operate as single-shop, manufacturing and selling units to meet the demand of the rural population. With a rich tradition of jewelry manufacturing, Pakistan's skilled/semi-skilled labor force is available at relatively lower rates, which offers a comparative advantage to the country. The cheap labor can be consumed at Gems and Jewelry Institute to generate overall revenue. [CITATION PGJ13 \l 1033]

There is also an opportunity to establish a gem identification and certification laboratory at the site to ensure better understanding of gemstones and their properties, perhaps in coordination with KU, NED or PCSIR and its gemstone Center of Excellence. The demand is increasing thus establishing a laboratory will help in boosting the revenues through added facility.

Even the Pakistani, Indian entrepreneurs aspire to strengthen ties with gems and jewelry when the delegate from India met with Pakistan that can boost the sales up to **\$ 200million initially**. [CITATION Amm12 \l 1033].

There are many institutes that are currently working on the system procedure of Gems and Jewelry institutes of Pakistan. India being the leader has around 6 major institutes, Bangladesh and Sri Lanka do not have any highly formalized bodies. Bhutan and Nepal lack in the facilities.

SMEDA has conducted a grand research on Gems and Jewelry industry that shows that due to the lack of supporting institutes Pakistan is standing at an annual \$30 million whereas India stand at \$10,589 million.

The major institutes in the ***SOUTH ASIAN REGION*** are and their facilities are discussed below,

GJEPC India

It has the State of the Art Institute equipped with the latest technology, paraphernalia of machinery gadgets, equipment and competent faculty resources make our education, deep and meaningful way to our jewelry trade and society as a whole. It is fully geared to cater to the need of the acute shortage of skilled manpower, faced by the Gem & Jewelry Industry. [CITATION Jai \l 1033]

Gemological Institute of India

It is a non-profit Public Charitable Trust established in the year 1971 by the Gem & Jewelry Exporters' Association. GII an independent and impartial institution is the pioneer in the field of Gemmology in the Country. Gemmological Institute of India in its journey since 1971 have kindled the light of knowledge in the field of Gemmology, Diamond Grading and Jewelry Designing amongst the gem trade and industry in India and also amongst the general public. GII is the only gem & diamond testing laboratory in Mumbai which is established by the Industry, for the promotion, propagation and development of the science of gemstones and research studies in India. [CITATION GII \l 1033]

IIGJ

It is the only full-fledged institute offering courses in all aspects of jewelry manufacturing. It has facility schools in New Delhi, Jaipur and Mumbai. IT provides teaching and practical teaching as well through their CAD labs and different state of the art equipment. [CITATION Placeholder1 \l 1033]

Curriculum Taught around the World in Major Training Centers

CANADIAN INSTITUTE OF GEMOLOGY (CIG)

Following are the courses taught at CIG:

GEM 110 GEMSTONES (Gemmology I)

This is a broad interest course presenting the basic classifying features of all significant gems, including their appearance, special physical and optical features, their occurrence and value factors. Also involved is an overview to diamond and colour gem grading, the use of gemmological instruments to aid identification.

Course Content:

1. Nature of Gemstones (1): Diamonds and Corundum 2. Nature of Gemstones (2): Beryl and organic gems 3. Crystal Structure and Origin of Gems, Opal, Topaz and Jade 4. Gem Cutting and Care: Garnet, Quartz and Spodumene 5. Gemstone Identification (1): Tourmaline, Chrysoberyl and others 6. Gemstone Identification (2), Diamond Grading (1) 7. Gemstone Economics: Markets and Values, Diamond Grading (2) 8. Colour Gem Grading and Appraisals 9. Gem Enhancement and Synthetic Gems 10. Review, Jewelry Manufacture, History and Fashion[CITATION Can13 \l 1033].

GEM 120 GEMMOLOGY II (Theory Course)

Refraction, reflection, gem crystallography, pleochroism, absorption spectra and other physical and optical properties of gems are covered in this course. It elucidates the numerous tools used in gem identification. The textbook “Gemmology, 3rd edition, 2007” by Peter G. Read follows standard exam requirements in this course.

Course Content: 1. Gemmology, the science of gem materials 2. Colour, lustre and sheen 3. Crystallography, the science of crystal structure 4. Hardness, fracture, cleavage and parting 5. Specific Gravity 6. Refractive Index and double refraction 7. The optics of polished gemstones 8. Absorption and emission spectra 9. Luminescent and electrical properties of gems 10. Organic gem materials and their simulants 11. Synthetic gemstones 12. Gemstone simulants.

GEM 140 JEWELRY, HISTORY AND DESIGN

This course is a necessity for all Jewelry collectors and fans. In ten lessons a review of Jewelry from the ancient world to the present day will be taken. Using the text-books as a visual guide students discover the varied styles, techniques and materials used from ancient Egypt, Greece and Rome to modern times showcasing the development of Western Jewelry[CITATION Can13 \l 1033].

GEM 150 GEM IDENTIFICATION

In twelve lessons students learn about pearl types and shapes including South Sea, black and freshwater pearls and how to judge luster, thickness and colour. History, occurrence, identification and the evaluation of jade and opal form the second part of this exceptional gemstones course.

GEM 160 PEARLS, JADE AND OPAL

In twelve lessons students learn about pearl types and shapes including South Sea, black and freshwater pearls and how to judge luster, thickness and colour. History, occurrence,

identification and the evaluation of jade and opal form the second part of this exceptional gemstones course.

THE COURSE POLICIES

After completion of all GEM 100 courses the student will receive the “Fine Jewelry Expert (C.I.G.)” wall certificate.

Students can start their course work at any time. Individual courses must be completed within 12 months; students may apply for extensions but may have to pay additional fees for new textbooks and study materials if necessary.

The passing mark for all written exams is 65%, for practical exams 100%. Students may attempt to re-write a failed exam two times after paying the required fee within 6 months (after a waiting period of 1 month).

In the Gem Identification course an organic liquid (methylene iodide) saturated with sulfur is being used in conjunction with the refractometer[CITATION Can13 \l 1033].

INDIAN INSTITUTE OF GEMS AND JEWELRY DELHI

DIAMONDS AND DIAMOND GRADING

Duration: 2 ½ months

The course provides comprehensive knowledge about Diamonds in all aspects. You will be confident to evaluate Diamonds on lines of the International Diamond Grading System and the famous 4Cs -Carat, Colour, Clarity & Cut. A balanced teaching theory & practical grading of actual diamonds will give a unique experience and a clear understanding of the product.

COLOURED GEMSTONE IDENTIFICATION

Duration: 3 ½ months

Student is taken into the fascinating world of gemstones by practical training on identification of a huge variety of real coloured gemstones from all over the world. The laboratory is equipped with professional gem testing instruments on which you will learn the identification of colored gemstones, distinguish between synthetic/imitation/treated and real gemstones and also learn about occurrence of gems, their inclusions and optical characteristics.

DIPLOMA IN GEMOLOGY

Duration: 6 months

This is a science, art and profession of identifying and evaluating of gemstones and diamonds. It is designed to open doors to the vast subject of Gemstones and is divided into two courses as under Diamonds and Diamond Grading Colored Gemstone Identification.

The above will round off the student's education in Diamonds & coloured stones to make him/her a real gemology professional. These courses can be availed individually or as comprehensive Diploma in Gemology [CITATION IIG14 \l 1033].

GEMOLOGICAL INSTITUTE OF AMERICA

GIA offers several programs and courses online through an interactive eLearning format, and through its 12 campus locations around the world. The institute also offers corporate training programs and works with trade organizations worldwide to provide technical training in gemstones and jewelry.

The Graduate Gemologist (G.G.) diploma offers a comprehensive education in gemology. Graduates of the program receive the Graduate Gemologist diploma as well as Graduate

Diamonds and Graduate Colored Stones diplomas[CITATION ACC11 \l 1033]. Students can also earn an Accredited Jewelry Professional diploma with the addition of one more course, which can also be taken independently[CITATION GIA11 \l 1033]. The Graduate Pearls diploma program provides a comprehensive foundation in pearl identification and grading.

Moreover, GIA's Carlsbad campus deals with several programs in jewelry arts. The Applied Jewelry Arts Program (AJA) diploma covers jewelry design, wax carving, mold making, casting and CAD/CAM. The Graduate Jeweler diploma program teaches the fabrication, repair and stone setting skills to become a professional bench jeweler. Other jewelry arts classes are held on campus in Carlsbad and New York.

GIA Diploma Programs and courses include:

- Graduate Gemologist
- Graduate Diamonds
- Graduate Colored Stones
- Graduate Pearls
- Accredited Jewelry Professional
- Jewelry Design & Technology
- Graduate Jeweler
- Jewelry Design Course
- Comprehensive CAD/CAM for Jewelry Course

Following Are The Tools Used By Gemologists Abroad During Training:

- 10X Loupe (a hand-held single, magnifying lens)
- Binocular Microscope (preferably stereoscopic)

- Polariscope (to test the dispersion of gemstones)
- Dichroscope (to test pleochroism of gemstones)
- UV Light (to test the luminescence of precious stones)
- Refractometer (to measure refraction of the stones)
- Spectroscope (to measure the stone's absorption of light)
- Specific Gravity Liquid (to distinguish between materials with similar characteristics)
- Weight Scales
- X-ray Photographs (to study x-ray diffraction in similar precious stones)
- Geiger Counter (to measure radioactivity of stones)
- Chelsea Filter (determines if a stone contains chromium and is natural or synthetic)
- Master Colored Grading Set (to grade the color of diamonds)

Career Opportunities for Gemologists

Some gemologists work as free-lancers, offering assessments for such things as insurance purposes or sales, others may also work in places as jewelry stores, auctions and gemstone laboratories, with jewelry outlets being one of the most prevalent places of employment for these professionals. However, jewelry stores, laboratories and auction houses are not the only places that gemologists may choose to work within. They may also work as buyers for jewelry outlets or as advisers for jewelry designers.

Many gemologists choose to work within museums, evaluating gem donations or advising on gemstone purchases. They will help the museum determine what gemstones that they should purchase or accept, and they will appraise them for insurance coverage. They will also determine what information the museum should include on any placards for precious stone exhibits.

Faculty Sources in Pakistan

Finding faculty which is willing to teach both theory and practical in Karachi won't that difficult as many educated people are already associated with this industry.

Following are the professionals we can approach in Pakistan for teaching at Pakistan Gems and Jewelry institute:

- Teachers and practitioners in Indus Valley School of Arts
- Entrepreneurs working in Gems and Jewelry sector
- Pakistan polytechnic college
- Pakistan Institute of Fashion and Design
- Students from local engineering universities willing to work part-time

International and Local Associations and Companies:

International:

Around the globe, the gems and jewelry business is being run with the support of gems and jewelry institutes and associations. Here is a list of a few such institutes and associations across the world, categorized by their geographical locations:

Asia:

- Indian Institute of Gems and Jewelry, Mumbai, India
- Gem Identification Course Institute, Delhi, India
- Gem & Jewelry Institute (GJI), Colombo, Sri Lanka
- Gemological Institute of Sri Lanka
- Asian Institute of Gemological Sciences, Bangkok, Thailand
- Gem and Jewelry Information Center, Thailand
- Gemlab Inc. (Thailand)
- Gemological Association of All Japan Co., Ltd.

Americas:

- American Gem Trade Association

- International Gem Society (IGS), USA
- International Colored Gemstone Association (ICA), New York, USA
- Gemological Institute of America (GIA)
- DeMello Gemological National Laboratory, Boston
- American Gem Society, Las Vegas
- Accredited Gemologists Association (AGA), San Diego, USA
- Canadian Institute of Gemology
- Canadian Gemological Association
- Brazilian Gems & Jewelry Trade Association
- Instituto Brasileiro de Gemas e Metais Preciosos (IBGM)
- Portuguese Gemological Laboratory, Portugal

Europe:

- International Gemological Institute of Europe, London, England
- Institut Gemmologique de France
- Centre de Recherche Gemmologique, University of Nantes, France
- Institute National de Gemmologie (ING), Paris, France
- Instituto Gemological Espanol (IGE)
- SSEF Swiss Gemological Institute
- Centre for Information & Services in Gemology (CISGEM), Italy
- German Gemological Association (DGemG)
- L'Ecole des Gemmes, France
- Gemlab (Europe), Athens, Greece
- Gemmes Inclusions, Paris
- The Gemological Association of Great Britain

- The Scottish Gemological Association, Scotland
- Swiss Gemological Laboratory

Oceania:

- The Gemological Association of Australia

Africa:

- Gemological Society of South Africa

Others:

- Russian Gemological Server, Moscow, Russia
- Alexandrite Gemstone Association
- International Gemological Institute (IGI)

Local:

Just like the rest of the world, Pakistan has its gems and jewelry business operative under government-linked associations, mainly. These associations and institutes include the All Pakistan Gem Merchant & Jewelry Association (APGMJA), Pakistan Gems and Jewelry Development Company (PGJDC), and Gems and Gemological Institute of Pakistan (GGIP).

Besides government support, these associations have several other stakeholders, too, whose list is available on the APGMJA website.[CITATION LIS14 \l 1033]

Human Resource:

Human resource development (HRD) is an ever-changing field. The concepts and the underlying principles of HRD may be similar throughout the globe and across all fields, but its practice differs due to contextual factors such as culture, technology, resources and national policies.

[CITATION Muh13 \l 1033]

To understand the context under focus i.e. Human resource development for the Gems and Jewelry Institute, Karachi, we need to categorize the institute by some general traits.

The institute under study can be broadly categorized as a small-sized training institute. That being said, we can research on the human resource development of similar institutes in the region, or across the globe.

The policy paper on HRD in TVET, Pakistan says that one of the most important factors in any educational institution or training school is the quality of the teaching faculty. The faculty, being the agent for imparting skills and learning, can create a dynamic of excellence or become tied to routine and mediocrity. It is the role of the faculty that determines the quality of students or trainees and the reputation of the institution. This is especially true for institutes where the students are not only required to be educated in a classroom environment but also required to be guided in actual work-shop conditions for hands-on training. [CITATION Pol14 \l 1033] The paper also emphasizes on the presence of only a limited number of training profiles and hence can provide training to only a small percentage of a fast growing young population.

[CITATION Sye07 \l 1033] emphasizes on the fact Pakistan's workforce is characterized as having low skills and poorly prepared to compete in today's globalized world. Rapid technological changes now require individuals to learn and relearn skills throughout their working lives by ensuring its relevance and effectiveness. For this purpose, the paper suggests that education and training require to be made need-oriented, multi-skill and flexible to meet changing needs of the local industry and overseas demand.

However, in our case, the human resource for the institute seems to be hard to be found, let alone be made competitive.

We can take the example of the Technical Vocational Education and Training (TVET) Sector in Bangladesh for which [CITATION MHT \l 1033] suggests public-private partnership (PPP) which can also be used as a solution for HRD in our case. Public-Private Partnership (PPP) is a model of development cooperation in which actors from the private sector (private corporations, corporate foundations, groups or associations of businesses) and the public sector (e.g. Ministry of Education and/or its agencies) bring together expertise and resources to achieve development goals. In a partnership, the public and private sectors jointly define goals, activities, roles, and responsibilities to be supported.

PRIMARY RESEARCH

Analysis of Surveys

Throughout the course of our project we used various qualitative research tools; mainly in-depth interviews and surveys. The surveys were conducted to determine both the input (students interested in learning) and output (jobs available for the learned students) potential in the market. Our target market was the jewelers of Karachi. We tried to focus more on the businesses involved in manufacturing as well as retailing. However, the jewelry market is majorly dominated by retailers since the manufacturing activities are usually outsourced. We conducted these surveys in the main jewelry markets of Karachi including areas of Saddar, Hyderi, Tariq Road, and Gulshan-e-Iqbal. (Check appendix for a sample of the survey questionnaire)**Exhibit 4**

The survey helped us gain valuable insights of the expectations and the demands in the market. Out of our respondents, almost all of the jewelers were focusing their business on gold and gold-imitation (artificial). Around half of these were also involved in silver or diamonds and gems. 100% of the respondents said that they would rather hire an experienced worker than a fresh-mind learned worker. Although this did not come as a shock, but it straightaway turns down the idea of expecting established jewelers to hire young, talented and learned workers over people who have been practicing these skills and been in the business for years. To learn about what kind of people get into the business, when we asked about where the labor usually learns the skills from, only 23% respondents said that they learn by themselves or (rarely) from an institute, while 70% respondents said that these skills are learned from within the family or by *ustaaads*. **(Exhibit 5)**

When asked about the areas the labor usually comes from the answers were mostly Saddar, Golimar, Hyderi, Dhorajee, Sachal, Lyari, and Garden. We tried to get a grasp of the skills in demand in the market so that we could recommend courses designed to teach those skills, for the institute. The most demanded skill is of designing while the least preferred skills are faceting and carving. Respondents also pointed out the significance of stone setting, polishing, and identification and grading. Jewelers also emphasized on how computer aided manufacturing and designing is far more efficient and convenient, and it is taking over manual designing by large. However, some traditional jewelers still prefer manual designing over software based designing. Majority of the respondents believed that there is a need to teach such courses in the country and they also appreciated the idea of a shuttle service, however, we rarely came across respondents who would hire trained and skilled labor from the institute. The foremost reason is just as it was

mentioned before; experience is an integral factor in such a field. Besides, manufacturers would never hire unknown labor until referred by someone due to trust issues and security reasons. All the different graphical analysis is done respectively in **Exhibit 6, Exhibit 7 and Exhibit 8**.

Proposed Courses:

We propose that the following courses be taught at the institute. Our proposal is based on the demand of skill-set by the jewelers, which we have found through market research. Evidence of why these particular skills are required to be taught, is also given below.

Gemstone Identification & Grading

This extensive course would cover both the theoretical and the practical part of gemstone identification and grading. The course would teach the students everything from the details and functioning of all the lab equipment involved, to the detailed identification of over forty popular gemstones. The course would also cover artificial treatments, coating, and differentiation between synthetic and natural gems.

The second part of the course would focus on teaching the students to grade and price gemstones accurately and consistently. It would also include knowledge on various colored stone grading systems and detailed but relevant information about each popular stone.

This would be 20 two-hour session course spread over a period of four months (1 two hour long class every week). The maximum recommended cap for the number of students enrolled in one class is 15, since this requires great concentration and focus by both, the teachers and the students. The eligibility criterion to be enrolled in this course would be Matriculation because basic know-how and understanding is however required. **Exhibit 9**

Manual Jewelry Designing

The basic objectives of this course would be to develop jewelry designing skills and educate the students with modern techniques used in jewelry designing ranging from pencil and paint brush handling to finishing. The course will help students create innovative, technically sound, and well calibrated designs through manual techniques. Furthermore, it would also help students maintain difference between replicas and original (inspired) designs. The course will also teach students render impressive presentations of their products and designs to the potential clients.

This course would comprise of 20 two-hour sessions spread over a period of four months (1 two hour long class every week). The maximum recommended cap for the number of students enrolled in one class is 15, since this requires great concentration and focus by both, the teachers and the students. The eligibility criterion to be enrolled in this course would be Matriculation because basic know-how and understanding is however required. It would be almost completely practical based. **Exhibit 10**

Computer Aided Jewelry Designing (CAD)

This comprehensive course on computer aided jewelry designing and manufacturing would equip students with skills necessary to become a CAD technician. It would facilitate them to design virtual 3D jewelry through licensed software (Rhinceros, Matrix, 3Design, ArtCAMJewelsmith, and Monarch), on-screen. Students will be able to create CAD models within the metrics of scale, proportion, and element relationships and also within the constraints of cost, time, size, style, and manufacturing methods. They will be taught to differentiate between the mentioned various CAD softwares and CAM methods. Similar to the manual designing course, students will be guided on how to model and render pieces that can be manufactured and display them in exhibitions.

The design of this course would be a little different since it requires a completely different set of learning instruments (computers, 3D printers...). The course would be spread over four months but would be divided into 40 one-hour sessions. Students would be given home assignments since they can practice this course at their homes as well. Recommended cap would be of 30 students. The class size would also cater the fact that this is the most demanded skill in the market. Students should hold certificate of Matriculation and should have know-how of computers. **Exhibit 11**

Stone Setting

This course would initially focus on the basics of stone setting; starting with basic hand engraving, simple shapes, basically teaching students the various styles and techniques of mounting gemstone on a piece of jewelry. Students would be taught the basics of how to prepare the jewel and calculate the stones, distances, and designs. They will be presented to different kinds of mathematical calculations. This involves measuring, drilling, and cleaning of various metal surfaces for setting. The course would later on focus on more complex and advanced

jewelry stone setting techniques. The course would include theory on different types of stones, cuts, shapes, settings, tools, and other specific techniques like star plate, bead and bright cut, bead thread work, pave layout, channel setting, etc.

This would be 20 two-hour session course spread over a period of four months (1 two hour long class every week). The maximum recommended cap for the number of students enrolled in one class is 15, since all work in this area requires extreme precision. **Exhibit 12**

Jewelry & Gemstone Cutting, Polishing & Finishing

This course would teach students these three vital skills that revolve around jewelry designing. The focus would be to teach the students the six basic cuts (Pear, Oval, Square, Octagon, Round and Marques), polishing, calibrating, and quality controlling. Students would be briefly made familiar with various carving and faceting machines, too. The techniques covered in the course would include the correct use of tools like files, hammers, etc., using a Dremel and pendant motors for finishing and polishing, using sanding and emery paper, texturing, and would also be taught to achieve mirror and satin finishes. **Exhibit 13**

Since the course teaches a variety of important skills, it is spread over three months but comprises of 15 one-hour sessions each month. The cap for the maximum number of students would be 15 so that one-on-one attention by the teachers is made possible while teaching these detailed skills.

View-point of the Association

The Association is firm on the point that the Institute needs to be revived to the basic stages and then expanded eventually. The basic courses that need to be taught in the initial stages, as identified by the Association will be related to the basic skills of Meenakari, Okaas, Safaai, Pardaas, etc. Other courses will be focusing on adding value to gemstones instead of gold, silver, or diamonds. The objective of these courses would be to teach students the basic skills of jewelry cutting, faceting, and polishing. The exact framework of the course-modules is yet to be finalized by the Association however there will be two separate programs for every course. The Basic Program would focus on a particular skill and will focus on teaching the basics of that skill in a period of two months. The Master Program would focus on a set of skills and its objective will

be that the student becomes eligible to deal in precious stones at the end of the two-year course. The mission of the Association is the well-being of the Pakistani gems and jewelry industry by injecting and grooming skillful labor, and their vision is to graduate hundred thousands of labor in the market in 5 years. For this reason they suggest that no fee is attached to the courses and all courses are taught free of cost. They also recommend that a stipend is set for the students who stick to learning after some period. At the completion of their respective courses, students will be exposed to opportunities in the industry as the Association has strong relations within. The students can then also opt to set up their own businesses (entrepreneurship).

As association's marketing idea also includes teaching basic technical courses of gems and jewelry in universities, especially government colleges/ universities, hence those students will also be facilitated if they are focusing on starting their own business related to gems and jewelry.

Mediums through which Courses will be marketed

The first part of our marketing strategy will be to develop collaborations with renowned universities. As our institute aims to provide international level education, we need to develop links with established institutes. Our proposal will be to establish a link between the Karachi University and our institute. KU does offer a variety of degrees to its students and we are going to offer their students free courses at our institutes. In return, KU is going to collaborate with us and market us to its students. Also similar deals will be signed with AIFD where their students will be provided facilities and in return they will sign a memorandum and market our institute. We will also keep in touch with these institutes regarding the demand of our courses. We will focus on what courses are demanded by their student and increase the total number of courses as time goes on. Collaboration with a university like KU is going to help establish our credibility; this is needed as we are a new institute and will require some time to establish ourselves. Our next focus is going to be collaborations with international universities. As our focus is on establishing an institute of international standards, this is an essential part. By collaborating with them we can learn a lot regarding new trends in gems and jewelry making, we can also share ideas regarding management, development of the gems sector etc. International institutes are keen on establishing links with Pakistan if they see genuine work being done here. We need to establish the name of the institute before we can collaborate with them. But once this is done, it

will be a huge boost for us, as students will want to come to the institute which has links internationally.

In addition to collaborations, personal selling will also be done to market our courses. We are going to approach different jewelers and pitch them the idea of sending their labor to our institute to learn. This is very important initially, as we need to increase the awareness level for our institute. We will also need to discover new jewelers who might also be interested in the idea of taking courses from our institute. This will be part of the persuasion effort where we convince the client to come to our institute.

As mentioned earlier, the training center's main target market is low income class. But for the incubator lower to middle class will be allowed. The whole purpose of this project is to facilitate the poor and hence rich people will not be our focus. Certain skills which are not very common around the world, especially hand work will be our focus. This is Pakistan's competitive advantage, even though most countries in the world are now completely machine-centric, Pakistan has lots of potential in hand made jewelry as well. Hence skills which are now starting to get redundant but are of immense value will be our focus at the institute. This can lead to great benefits for Pakistan in the future.

In addition to all these, we are going to advertise in newspapers. That is essential to reach out to a large audience. We can advertise once every few months in reputable newspapers to create awareness amongst people regarding our institute. Most of the advertising efforts will be done by the institute in the starting months of the years i.e. Jan- May, as following these months are the summer holidays of students in which they can join the institute in great number. So, it's essential to spread the word amongst the target student body and the masses before the summer holidays.

Different pamphlets will be printed, and distributed throughout Karachi, in markets like Hyderi, Saddar, Tariq road so as to ensure jewelers know of our institute. These pamphlets will also be a part of the marketing campaign of the government schools.

Through all these marketing techniques we are going to create awareness and attract customers to our institute.

Machineries

The machinery already at hand is:

Grinder: This machine is used to break the stones into small pieces so that they may be used according to their size requirement.

Cutting Machine: Used to cut gems and jewelry into smaller pieces which are otherwise very hard to cut.

Vibrators: Used for polishing the jewelry. The jewelry loses its shine with the passage of time and to get the shine back of the jewelry, this machinery is used.

Beading Machine: They are ideal for doing edging and detail work on sheet metal panels.

Lath Machine: Is a machine tool which rotates the work piece on its axis to perform various operations such as cutting, sanding, knurling, drilling, or deformation, facing, turning, with tools that are applied to the work-piece to create an object which has symmetry about an axis of rotation.

Faceting Machine: The faceting machine is used to rub and level the surface of the gems. Faceting is the process of creating the required faces/surfaces of the gem and the price of the gem is directly proportional to the quality of the faces.

Machinery to be acquired:

Following are the machineries which we believe must be acquired to make the operations of the incubator and the training institute relevant to the market demand. These machineries are basically being recommended after thorough market research and online secondary research. We have consulted every party involved in the supply chain of this institute i.e. manufacturers, wholesalers and retailers, regarding the skills and the machineries most used in the market right now and what will be demanded in the future.

The idea of operating the institute is to run it as a training institute in the morning and as an incubation facility in the evening. The machinery would be installed according to the demands of both type of institutions we are planning to run (i.e. training institute and incubation facility).

The training institute will be running on light machinery as the focus of training provided would be on basic skills; skills which are dying in the market but are highly demanded in international market. For the incubation facility, we would be acquiring some of the heavy machinery (CAD/CAM etc.) as only from those machinery we would be able to attract the jewelers from the market who don't already possess them. Below is the list recommend for the institute to acquire over a period of time to maintain its competitiveness with the market:

Name	Quantity at hand	Quantity to be acquired	Cost of one piece (Rs.)	Description	Recommended for:
Distillation Unit	0	5	70,000	Used for separating impurities	Training Institute
Gemological Microscope	1	3	760,000	Used for identification	Incubator
Carving Kit	0	5	120,000	Used in the process of gems and jewelry carving	Training Institute
Diamond Tester	0	5	190,000	To test whether the diamond is real or not. Also used in diamond grading.	Incubator
Refractometer	0	5	200,000	Used for the measurement of an index of refraction	Incubator
Computers (Pentium 4)	1	7	7,000	For CAD software and rapid prototyping	Incubator
CAD software (Rhino)	0	1	150,000	-	Incubator
3D Printer (SolidScape R66)	0	1	30,000,000	-	Incubator
Cutting and Polishing machine for thinner section	0	1	6,000,000	For cutting and polishing at a very fine level	Incubator
Vibrasonic Tumbler	1	2	50,000	For tumbling gems and jewelry	Training Institute
Hard Rock Trimmer	0	1	100,000	For trimming hard	Training

				rocks	Institute
Faceting Unit	1	9	200,000	Faceting of gems and jewelry	Training Institute
Gold Plating machine	0	1	900,000	For using electrolysis in gold polishing	Incubator
Gold/Silver spot welding machine	0	1	400,000	For welding in the bangles making and other jewelry processes	Training Institute

Furniture and Fixture:

Item	Quantity at hand	Quantity required	Per unit cost (PKR)	Total (PKR)
Work bench/Tables	12	0	-	-
Chairs	50	0	-	-
Steel Amirah	1	2	20,000	40,000
Split A/C	1	0	-	-
Total				40000

View-point of the Association

The Association believes that only start-up investment is required to acquire basic machinery. It is certain that there is a need for basic skills in the market at the moment. The model it is adapting is to ignite a spark with minimal investment and once it turns into a huge fire, additional investment may or may not be required. The Association has the point of view that even CAD and CAM are mostly required for diamond jewelry and Pakistan does not deal in diamonds, majorly. Instead there is a need to add value to gemstones. For this, basic machines for gem cutting and faceting, etc. are required. Some of these machines are already present at the institute but they are mostly in deplorable condition. So, investment is required to get those machines in operating condition and maintain them. The machines that are not available but are required to teach these basic skills are to be imported from within the continent, instead of getting them from overseas. This would save lots of money, according to the Association. It further believes that manual designing will always remain worthy in spite of whatever technological changes take

over the industry. This supports the argument that only basic machinery is required to teach the basic skills, and the same skills are in demand.

Marketing of the incubator

The business incubator is an essential part of our plan which is to enable the institute to sustain itself. The incubator is going to benefit the young entrepreneurs who have just started their business and need a place and the required machinery to work on gems and jewelry. There are many people who do not have required resources and hence are not able to advance in this industry. We focus on helping them expand their business by giving them the appropriate resources for their work.

The target market of the incubator is very different from the institute. Here our main focus is young entrepreneurs/ businessman. Hence our marketing means are going to be very different. We are going to focus on social media. We will develop a Facebook page where we will have all the details of our incubator and what it offers to people. The page will be constantly updated and will answer all questions regarding the incubator. We will also be developing our website so as to give people as much as information as possible because people involved in the jewelry business need to ensure the trustworthiness of the organization. Also blogs will be written regarding the incubator highlighting its advantages to businessman. Through all these, we aim to increase awareness of the incubator as it is a completely new idea and we need to market it to the relevant audience.

Another marketing technique we will use is personal selling. As gems and jewelry business is a matter of trust and confidence, we need to approach the customers directly. We will explore the market, to identify new entrepreneurs who need our facilities. We will pitch in our whole concept of business incubator and how we have the necessary machinery, equipment and land which can provide the perfect environment for new and budding entrepreneurs. There are many small scaled businesses that require such facilities according to research that we have done, and we believe many are going to be interested by this idea.

Apart from that, we are going to advertise in the newspaper. Newspapers are the best way to communicate your message to a larger audience. Again, we do not have that large a budget hence

we will only advertise in the starting few months. This again is to increase our awareness amongst the target audience who are constantly looking for the kind of opportunity we are providing.

Hence through these combine methods; we are going to market our institute/incubator.

View-point of the Association

In Pakistan, it is not very common for people to go to institutes to learn skills related to gems and jewelry. Instead the focus is more on self-learning. The Pakistan Gems and Jewelry Institute strives to change that. It aims to develop such a course outline that people who study here learn skills in an environment similar to how it is taught internationally.

Target Market

The Pakistan Gems and Jewelry Institute was founded in 1998 with the collaboration of All Pakistan Gems and Jewelry Association. Its aim was always to target the lower class, those who can't afford to study or take even basic courses related to jewelry. The Pakistani market has huge potential, but it has been unable to fulfill that potential mainly due to the lack of availability of skilled labor. The institute's main aim has always been to train unskilled labor, teaching them various skills which are not very common and the lack of which has left Pakistan missing out on fulfilling its potential. Hence APGMJA wants to focus solely on the lower class, and it will hire those few skilled people present in Pakistan to teach others these skills. Pakistan has abundance of gems present but it has been unable to profit from it due to lack of value addition. With this institute, this problem can be addressed. APGMJA is especially focusing on training young girls. Traditionally, women have been involved in this sector for years, and APGMJA will train young girls, teaching them all the required skills. The institute in the start is going to offer basic courses in gems and jewelry making. The educational requirement will be Matriculation. Later on APGMJA hopes that it can introduce more advanced courses as well. The state of the art equipment and laboratories will help the Pakistani gems and jewelry immensely which is dire need of an institute that is in line with modern technology. With its focus on modern technology and skill development as well as ensuring jobs for its students, the institute has the potential for being a huge market for people involved or interested in the Gems and Jewelry sector.

The APGMJA is focusing on an innovating idea for the institute. Their aim is to teach gems, jewelry making skills to prisoners. They are following the example of China, which has also taken this initiative. Through this, thousands of people can be trained in the next 5 years and Pakistani gems and jewelry industry which lacks unskilled can receive a major boost.

Another marketing strategy of APGMJA is to collaborate with various technical universities of Pakistan to teach gems and jewelry institutes. There are a few institutes which do teach gems and jewelry basic courses. But APGMJA is trying to introduce basic courses of gems and jewelry making in these technical institutes by collaborating with them. Hence, universities like KU, Indus Valley School where students will be willing to take courses related to gems would introduce these courses and can send their students to take further courses from the institute for free. This is going to be a major boost for the sector with many students getting technical skills of gem making. Collaboration with a university like KU is also going to help establish the institute's credibility; this is needed as it is a new institute and will require some time to establish itself.

Hence if all these plans can be implemented successfully, the Pakistani gems and jewelry industry can soon be on its feet and Pakistan's exports in this sector can increase manifold.

Funding Proposal

For funding, we have developed a sample proposal that shall be used for fund-raising purposes.

Introduction:

The ELP is an integral part of the BBA program offered at the institute of business administration, Karachi. A group of 4-5 final semester students undertake a management consulting assignment in real life business environment related to managerial and organizational problems that need solutions. The projects not only enable students to develop

interviewing and report writing skills, but also provide an opportunity for them to enhance their managerial and leadership skills, besides their problem-solving and decision-making skills.

Project Detail:

Our project is with the Trade Development Authority of Pakistan (TDAP). The main objective is to revive the Pakistan Gems and Jewelry institute which was formed in 1999. Pakistan Gems and Jewelry Institute is basically a training and development institute, which was founded to train the labor in the skill-set of gems and jewelry. The project has been closed for the last 9 years and heavy capital has been invested in it which is actually being wasted there. Our job is to correctly analyze the current scenario of the institute and to present a business plan (for sustainability) and a proper curriculum and SOPs for training purposes of the institute.

This project is under Mr. Haider Zamin (Assistant Director – Export Development Fund) for TDAP and the faculty advisor following our progress will be Mr. Shahid Mir.

What activities does TDAP regularly undertake?

- *Participates in 60 to 80 international trade exhibitions annually.*
- *Sends 20 to 40 trade delegations abroad every year.*
- *Organizes the EXPO PAKISTAN annually in Pakistan, which is well-attended by foreign buyers.*
- *Runs the Expo Centre, Karachi providing a permanent exhibition space for holding trade events.*
- *Implements various Trade Policy Initiatives announced by the Commerce Ministry.*
- *Undertakes various sector-development projects from the Export Development Fund.*

Market of these institutes in Pakistan:

The revival of the Gems and Jewelry Institute is very vital for the Pakistani industry. The gems and jewelry industry has huge potential and it has grown phenomenally in the last few years.

Having been blessed with a variety of gemstones, and being one of the top ten consumers of gold in the world, the industry holds great potential to attract investment in the sector. Hence it is vital that skilled labor is available in the industry and that is why the revival of the institute is so essential. There are many people in the industry that would prefer taking proper courses to learn about gems and jewelry.

Currently most people in the industry have learnt the art without any professional training. It usually runs in the family. The target market for this institute would be those people who are either in the industry already or those who want to join it. Young people of age 18-25 would be our main target market. Education requirement will be Matriculation for most basic courses, but some courses which are advanced will require advanced technical education. The institute is a project by the Government of Pakistan, and its main aim is to cater to those who cannot afford to learn gems and jewelry processing. They have to learn the art from their fathers or by any other means. Apart from that, people who have been involved with the industry, whether they are businessmen or employees, they can also get training in these institutes. Many business graduates who do not have technical skills but own jewelry or gem related businesses, can also take courses so as to learn about jewelry making from the institute. The institute is in Karachi and hence it will target the Karachi market. Karachi is the economic capital of the country, the largest city and therefore it has a large market. It also has a huge rate of unemployment and with the revival of this institute more people can learn technical skills which will help them get employment. Also, many young entrepreneurs are now in the gems and jewelry making business; the institute will provide an ideal opportunity for them to learn as the institute will have the latest machines and technology similar to what is available in the West. The state of the art equipment and laboratories will help the Pakistani gems and jewelry immensely which is dire need of an institute that is in line with modern technology. With its focus on modern technology and skill development as well as ensuring jobs for its students, the institute has the potential for being a huge market for people involved or interested in the Gems and Jewelry sector.

Marketing of the institute will be done mostly through personal selling, meeting with Jewelers and other people involved with the industry. Social media will also be used to attract people to our business incubator.

Statement of need:

What we are proposing is forming a business incubator at the training institute. This will cater to entrepreneurs who want to enter the jewelry industry but do not have the required the sophisticated machinery to make products. The machine and land will belong to the institute and labor, expertise will of the entrepreneurs. As far as training center is concerned, we are going to offer courses like

- *Gemstone Identification & Grading: Students should be able to identify gemstones using various gemological instruments and should have basic understanding of gemstone grading*
- *Diamond Grading: Students should be able to grade diamonds according to international standards*
- *Gemstone Faceting: Students should master the art and engineering of mechanically cutting facets on pieces of gemstones*
- *Gemstone Carving: Students should master the art of gemstone carving, to develop different objects and enhance the beauty of gemstones*
- *Manual Jewelry Designing: Students should be able to create innovative and technically sound designs, manually*
- *Computer Aided Jewelry Designing & Manufacturing (CAD/CAM): Students should be able to design virtual 3D jewelry on-screen, using software like AutoCAD*
- *Wax Carving: Students should be able to transform creative ideas into a practical model by making patterns in wax form*
- *Jewelry Finishing & Polishing: Students should have learnt the techniques of jewelry finishing and polishing*

- *Stone Setting: Students should master various styles and techniques of mounting gemstones on jewelry*
- *PATWA: Students should master the art of stringing and hanging beads and pearls on jewelry*
- *SOU MAK: Students should master the weaving techniques of making jewelry by using wires and beads*
- *Jewelry Marketing: Students should have learnt the marketing theories and methods applied to jewelry markets and should be able to render impressive presentations to clients*
- *Cost Management: Students should be able to use cost accounting to support decision-making*

For all these courses, we will require teachers who are skilled in this art as well as sophisticated machinery. The institute does have simple machinery, but if it is to turn into a respectable institute it needs better machinery. Also, the business incubator also requires better/ more sophisticated machinery. On the other hand, as the institute has been closed now for many years, not many people are aware of it. Hence it has to be marketed to the students/ jewelers as well as other institutes. For that, we require heavy initial investment and for that we would really appreciate USAID assistance. Below are the details of the current available resources as well as required resources and the money required for them.

Available resources:

Currently the building of Pakistan Gems and Jewelry is located in Defence Phase 2, Karachi. It is built on an area of 120 and has three floors. The machines that are available at the institute include various simple as well as complicated machines for processing, shaping of jewelry and gems like:

Silicon Carbide Sanding belt, Flame photometer, Roma micro spectrometer, Analytical balance, Conductometer, Gravity meter, Shaking table, Distillation unit, Fluid inclusion stage, Metal

detector, XRF, High temperature furnace, Ultrasonic water bath, Gemological microscope, Bausch and lamp loup, Belt sander, Bead mills, Centrifuge, Calcite dichroscope, Diamond tester, Gems pro refractometer, Grinding kit, Heart and arrows scope, Melting point apparatus, Hannerman gem filter set, Muffle furnace, OPL spectroscope, Polishing heads face, Polishing medis, Diamond colorimeter, Polariscope, Gem carat scale, Slab saw, Trim saw, Calibrating machine, Ultrasonic drilling machine, Gem carving machine, Barrel finishing machine, Vibrasonic tumbler, Dual grinder, Expendable drum combination, Sphere making machine, Automatic cabochon machine, Lath machine, Fordom flexible shaft machine, Lath machine, Sand blaster, Diamond discs, Copper polishing laps, Tin polishing laps, Steel polishing laps, Spacer, Vinyl polishing lap, Carving tools, Sponge rubber laps, True circle diamond disc, Diamond hallow drills, Leather polishing belt, Solder flux, Pressure Casting Machine, Vacuum Casting Induction, Portable gold melting machine, Graphite Crucible, LASER Welding and Marking.

Needed Resources:

We would like to have USAID on board majorly for the up gradation of the Institute to become competitive as compared to the private factories in the market.

To improve the effectiveness of the Institute we would like to impose a CAD technology.3D printing & Milling machine.

CAD/CAM Solutions:

Wax Model Making

*This model making solution comes complete with powerful design software that allows you to create detailed 3D models that clients can review from every angle. From these files, the 4 axis desktop mill automatically produces intricate wax models of beautifully crafted rings, pendants, bracelets, brooches, earrings and cufflinks. **Exhibit 14***

Cost: 2000 USD per piece, minimum of 3 pieces are required.

Personalized Pendants and Bling

*For an unbelievably low entry-price, you can add value to a wide range of jewelry and gifts, turning them into treasured mementos. **MPX Photo Impact Printers** quickly and easily add text,*

photos, logos and other artwork to a wide variety of metal and acrylic items from souvenirs and plaques to charms and pendants. **Exhibit 15**

Cost: 13000 USD per piece. Minimum of 3 pieces are required.

MDX-40A CNC Mill/CNC Machine

This subtractive rapid prototyping technology creates accurate prototypes for complex snap-fit and proves part manufacturability better than additive methods. It also handles a wide variety of materials and generates a smooth surface finish. CNC mill, bundled with 4-axis SRP Player CAM software makes it quick and easy to go from a CAD model to physical part or prototype. Large 12" by 12" by 4.1" maximum work volume without rotary axis and an optional rotary axis supports materials up to 10.6" long by 4.7" in diameter. **Exhibit 16**

Cost: 8000 USD, minimum of 3 pieces.

Besides these three machines we also look forward for your collaboration in the courses we will offer at the institute, if USAID can provide our students with different opportunities to learn different short term courses outside Pakistan in US. Along with conducting different courses in Pakistan from foreign faculty.

Item	Price per piece	Total Quantity	Total Price
Wax Model Making	2000 USD	3	6000 USD
MPX Photo Impact Printers	13000 USD	3	39000 USD
MDX-40A CNC Mill/CNC Machine	8000 USD	3	24000 USD
Miscellaneous	USD 6000		6000 USD
Total			75000 USD
Total			7.5 million PKR

Total Budget:

Costing

Capital Expenditure		Description	Quantity	Per-unit cost	Total Cost
	Machinery				
		Distillation Unit	5	70,000	PKR 350,000
		Gemological Microscope	3	760,000	PKR 2,280,000
		Carving Kit	5	120,000	PKR 600,000
		Diamond Tester	5	190,000	PKR 950,000
		Refractometer	5	200,000	PKR 1,000,000
		3D Printer (Solidscape R66)	1	20,000,000	PKR 20,000,000
		Cutting and Polishing machine for thinner section	1	6,000,000	PKR 6,000,000
		Vibrasonic Tumbler	2	50,000	PKR 100,000
		Hard Rock Trimmer	1	100,000	PKR 100,000
		Faceting Unit	9	200,000	PKR 1,800,000
		Gold Plating machine	1	900,000	PKR 900,000
		Gold/Silver spot welding machine	1	400,000	PKR 400,000
	Furnishing				
		Work bench/Tables	0		
		Chairs	0		
		Steel Amirah	4	20,000	PKR 80,000
		Split A/C	6	56,000	PKR 336,000
	Computer				
		Core i3 desktops	10	20,000	PKR 200,000
		CAD Software Subscription	1	150,000	PKR 150,000
	Total Capital Expense				PKR 35,246,000

	Description	First Year	Second Year	Third Year	Fourth Year	Fifth Year
Machinery						
	Distillation Unit	PKR 140,000	PKR 70,000	PKR 70,000	PKR 70,000	PKR -
	Gemological Microscope	PKR -	PKR 760,000	PKR 760,000	PKR -	PKR 760,000
	Carving Kit	PKR 120,000	PKR 120,000	PKR -	PKR 240,000	PKR 120,000
	Diamond Tester	PKR 380,000	PKR 190,000	PKR 190,000	PKR 190,000	PKR -
	Refractometer	PKR 400,000	PKR 200,000	PKR 200,000	PKR 200,000	PKR -
	3D Printer (SolidScape R66)				PKR 20,000,000	
	Cutting and Polishing machine for thinner section	PKR -	PKR -	PKR 6,000,000	PKR -	PKR -
	Vibrasonic Tumbler	PKR 50,000	PKR 50,000	PKR -	PKR -	PKR -
	Hard Rock Trimmer	PKR -	PKR 100,000	PKR -	PKR -	PKR -
	Faceting Unit	PKR 800,000	PKR 200,000	PKR 400,000	PKR 400,000	PKR -
	Gold Plating machine	PKR -	PKR -	PKR -	PKR -	PKR 900,000
	Gold/Silver spot welding machine	PKR -	PKR -	PKR 400,000	PKR -	PKR -
Furnishing						
	Work bench/Tables					
	Chairs					
	Steel Amirah	PKR 80,000				
	Split A/C	PKR 112,000	PKR 112,000	PKR 112,000	PKR -	PKR -
Computer						
	Core i3 desktops	PKR 80,000	PKR 60,000	PKR 40,000	PKR 20,000	PKR -
	CAD Software Subscription	PKR 150,000	PKR -	PKR -	PKR -	PKR -
Total Capital Expense		PKR 2,312,000	PKR 1,862,000	PKR 8,172,000	PKR 21,120,000	PKR 1,780,000

Projections

Projected Income Statement Pakistan Gems and Jewelry Institute						
		1st year	2nd year	3rd year	4th year	5th year
Net Sales (PKR)		PKR 891,359	PKR 2,614,388	PKR 13,823,706	PKR 46,035,283	PKR 48,677,526
Expenses						
Training Institute						
	Marketing & Promotion	PKR 170,000	PKR 170,000	PKR 205,000	PKR 175,000	PKR 170,000
	Office Expense (postage)	PKR 7,096	PKR 9,823	PKR 13,597	PKR 18,821	PKR 26,053
	Office Supplies	PKR 14,192	PKR 19,645	PKR 27,193	PKR 37,642	PKR 52,105
	Trainer's Salaries	PKR 840,000	PKR 840,000	PKR 1,260,000	PKR 1,260,000	PKR 1,260,000
	Stipend to students	PKR 28,384	PKR 39,290	PKR 54,387	PKR 75,284	PKR 104,211
	Miscellaneous	PKR 11,354	PKR 14,192	PKR 19,645	PKR 27,193	PKR 37,642
Incubation Facility						
	Marketing and Promotion	PKR 30,000	PKR 30,000	PKR 30,000	PKR 30,000	PKR 30,000
	Gold dust expense	PKR -	PKR -	PKR 1,372,541	PKR 477,514	PKR 477,514
General Expenses						
	Oil Expense	PKR 24,000	PKR 36,000	PKR 36,000	PKR 48,000	PKR 48,000
	Machinery maintainence	PKR 89,136	PKR 261,439	PKR 1,382,371	PKR 4,603,528	PKR 4,867,753
	Admin staff Salary	PKR 480,000	PKR 480,000	PKR 1,080,000	PKR 1,080,000	PKR 1,080,000
	Janitorial Staff Salary	PKR 84,000	PKR 84,000	PKR 84,000	PKR 84,000	PKR 84,000
	Security Guard Salary	PKR 360,000	PKR 360,000	PKR 540,000	PKR 540,000	PKR 540,000
	Utilities	PKR 1,387,200	PKR 1,117,200	PKR 4,903,200	PKR 12,672,000	PKR 1,068,000
Total Expense		PKR 3,525,362	PKR 3,461,589	PKR 11,007,934	PKR 21,128,983	PKR 9,845,277
Net Income		-PKR 2,634,003	-PKR 847,201	PKR 2,815,772	PKR 24,906,300	PKR 38,832,248

	Financial Ratios									
	Pakistan Gems and Jewelry Institute, Karachi									
						1st Year	2nd Year	3rd Year	4th Year	5th Year
Ratios										
Revenue per employee	Revenue					PKR 127,337	PKR 373,484	PKR 1,382,371	PKR 4,603,528	PKR 4,867,753
	No. of Employees									
Return on Capital Employed	EBIT					-113.93%	-45.50%	34.46%	117.93%	2181.59%
	Capital Employed									
Net Margin	Net Income					-295.50%	-32.41%	20.37%	54.10%	79.77%
	Net Sales									
Operating Expense Ratio	Operating Expense					395.50%	132.41%	79.63%	45.90%	20.23%
	Total Revenue									
Revenue Growth	Current period - Previous period sales					0.00%	193.30%	428.75%	233.02%	5.74%
	Previous Period sales									

Potential Clients

The gem and jewelry industry of Pakistan is highly influenced by the family traditions that are continuously being passed over to generations without bringing in any considerable transformation. This has led to a creation of hegemony of a few families and hence blocking way for new potential entrants in this segment of the industry. This hypothesis, obtained from our primary as well as secondary research, along with other arguments is well indicative of the fact that the institute we plan to revive is going to face some serious issues in attracting the appropriate clientele that would suit the profile of a “training institute”.

Therefore it is imperative for the concerned organization to look into the options under review with immediate attention being bestowed to the relevant target market we plan to co-ordinate with and hence explore additional uses of the existing facility. Perpetuating the options being considered we have worked on to bring the following analysis to them.

Under the first option, we have planned to set up an incubator in-collaboration with one of the relevant organizations with the primary objective of setting-up and growing entrepreneurial ventures in the gems and jewelry industrial sector of Pakistan along with necessary support and financial and technical services. The acquiring organization will be primarily responsible to provide a wide array of business support services and resources that could include but not limited to physical space, initial funding capital, management training, networking connections, etc.

The incubator is expected to offer several benefits that the entrepreneurs could avail and hence contribute to the development of this naïve industry. This spectrum is going to cover activities ranging from providing office and manufacturing facilities at a subsidized rate, which is going to be well below the existing market rates, to providing the highly demanded expertise in developing and implement business plans.

This incubator will pool in resources for different businesses, which will operate under its umbrella. By doing so, the incubator shall aim to achieve economies of scale as well as economies of scope to a considerable extent. Such an initiative will divide a majority of operational and overhead costs across those businesses, which will eventually result in the lowering down of these costs for each of the individual units employed there.

We are looking into possibilities for partnering with the Centre for Entrepreneurial Development at IBA, Karachi or the LUMS Centre for Entrepreneurship. These centers have an established reputation in providing incubation services to various businesses across several different industries all over Pakistan. Apart from this vast array of experience these centers have the ability to pool in resources with such a well-qualified faculty at their disposal. With such skills we believe that this incubator will serve as the starting switch of a modernizing revolution that will take up this entire industry. The responsibility of bringing in those businesses will lie with the overtaking authority and a TDAP representative will also have a representation in the key issues and the future path for this incubation initiative.

The second feasible option, which we consider to be the fate of this institution, is to shut it down and let some other institution to take it over and explore alternate uses of this facility. The details that support this argument have already been provided in the subsequent sections of this report and we believe that the sole purpose of this institution will have to be revisited and hence we suggest a change in the core strategy of this facility. Some of alternate uses of this facility could be to sell this institution of PJGDC, as they already have a well-functioning training institute in Saddar, Karachi and therefore they are high on learning curve to expand their reach and serve the ideology for which this institution was initially set up.

Apart from this a jewelry workshop for the manufacturing of jewelry could be set up and this facility could be open to various jewelers to use it to perform their necessary tasks. We believe that this idea is again not supported by the core purpose of this institution and hence it cannot be implemented for as long as this facility remains under the umbrella of TDAP as a training institute.

Conclusion

The purpose of this final report is to present 3 simple options in front of TDAP (Trade development authority of Pakistan) for running the institute. The secondary research previously done is also attached with this report for the reference purpose and the primary research data and results are also attached here. The main purpose of this report would be to give recommendations to TDAP which are derived after thorough primary research and an in-depth analysis of the collected data.

The recommendations to run the institute are following:

- Business Incubator Facility
- Public Private Partnership
- Training Institute

The concept of a business incubator is to provide space and equipment to business owners (especially small entrepreneurs) who cannot afford to make their own workshop so that they may utilize the facility to make their required products by giving a nominal fees. This concept is introduced to primarily make the institute self-sustainable (not to earn profit) as the cash flow obtained from the government sources is not reliable. There are two ways through which this concept can be incorporated in this institute:

1. Only Business Incubation Facility
2. As part of training institute

The first option is basically to shut down the training institute and operate the facility as a business incubation facility. This option considers the assumption that the location of the institute is not viable to run a training facility and the institute won't get much response when it comes to number of students coming for training. This option is basically just to make a good use of the facility and to prevent it from being wasted away. The second option is more viable and can be used in a way that the students in the training institute are getting hands-on experience working with the outside businessman coming to the facility for using the incubation facility of the institute.

The primary organization we would recommend for a Public Private Partnership is the “All Pakistan Gems and Jewelers Merchant Association”. The reason for recommending this organization for a partnership is primarily to take advantage of their vast experience, in-depth knowledge and contacts in the Gems and Jewelry sector. The association was previously also involved in running the same institute 15 years back but due to some internal matters of both the organizations (i.e. TDAP and APGJMA), the institute was unable to become operational.

But now, APGJMA has a long term growth plan for the institute and is very much interested in running the institute as the institute would be providing their workshops and other manufacturing unit with skilled labor.

Another avenue which can be looked for the public private partnership is the partnerships with other training and educational institutes of the country in which the institute will be run under the management of other institute and the control will remain with TDAP. The challenge with this model of partnership is that no other party will be ready to overtake the institute until they see some potential in it. For that, it is essential that the institute must be operated by TDAP for some time and then later when it comes upon the stage that the feasibility of the institute can be proved to other organization then this idea can be pitched to them. Potential organizations for this kind of partnership are Institute of Business Administration (IBA) Karachi and Pakistan Gems and Jewelry Development Company.

The last option is to simply operate the training institute using the resource of TDAP and it will then only serve the purpose of a training institute. According to our study, following are the issues associated with this option:

- No self-sustainability involved
- Location of the institute will make it hard for the target audience to approach. As major hubs of the manufacturing facilities are located very far from the site of the institute.
- A already well equipped training institute of gems and jewelry by the name of Pakistan Gems and Jewelry Development Company (PGJDC) is located in Sadar (a hub of jewelry manufacturing) but from our visit to that institute we came to know that the student's turnover is very low there.
- The increasing gap in the demand and supply of the gems and jewelry manufacturing labor is a major cause of concern as increasing inflation has adversely affected the

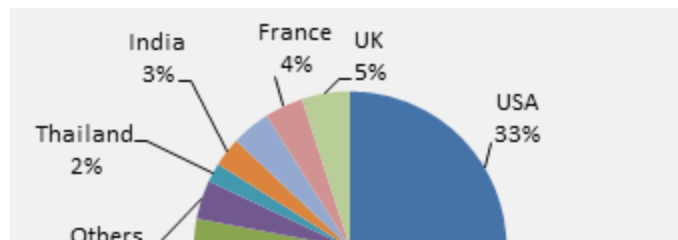
aggregate demand and this is a major negative effect on the luxury goods which includes the gems and jewelry.

APPENDIX

Exhibit 1: Global Distribution of Gold and Gems

Exhibit 2: Jewelry

Exhibit 3: Gem



Making Process

Cutting Process

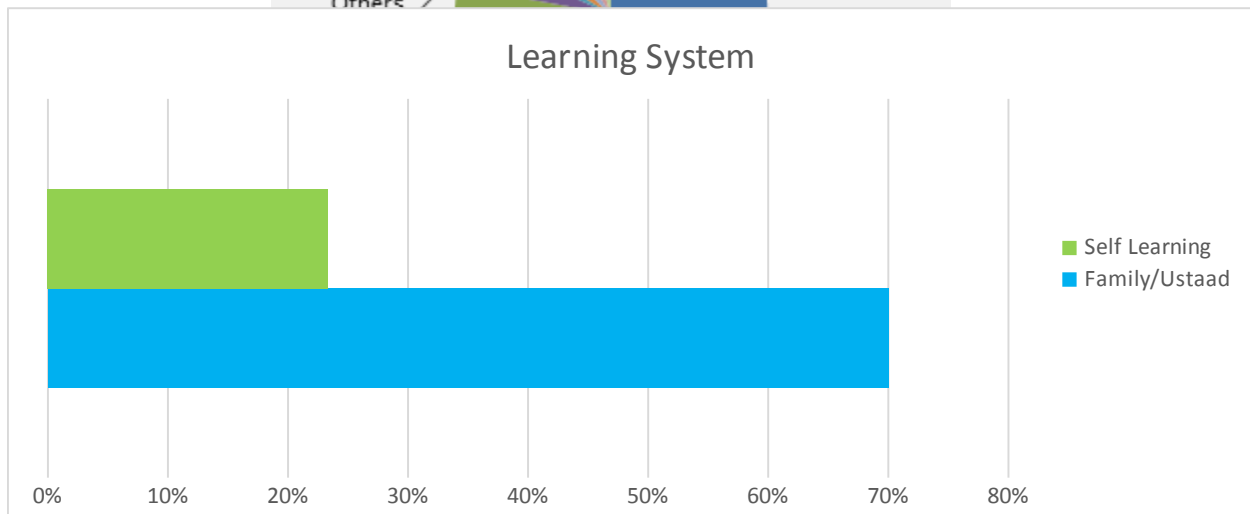


Exhibit 5: Learning Methodology

System	By Percentage
Family/Ustaad	70%
Self-Learning	23%

Sawing

Grinding

The object
finished pro

Sanding

Lapping

Karachi

Step 2:

After the wax sculpture is completed it is then encased in a silica encasement or in the case of jewelry it is then

Polishing

Step 3:

Then after the plaster encasement has hardened and dried it is put into an oven or kiln. This will cause the wax to melt and burn out leaving its hollow image inside of the plaster encasement.

Step 4:

Then the plaster encasement with the hollow shape of the piece of jewelry is placed into a casting centrifuge which is a device that will use centrifugal force to drive molten gold down through an opening in the chunk of plaster where it will fill the hollow spot left by the wax melting out.

Step 5:

After it has cooled the plaster is broken open to reveal a piece of gold jewelry where the wax sculpture once was.

Business

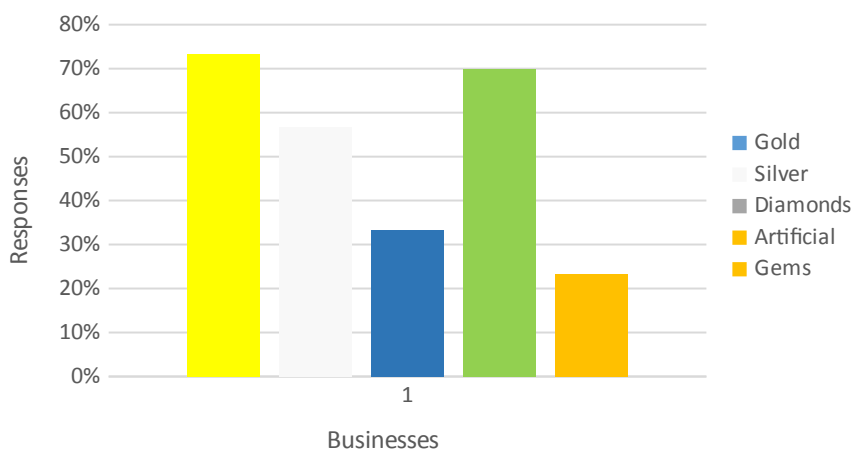


Exhibit 7:

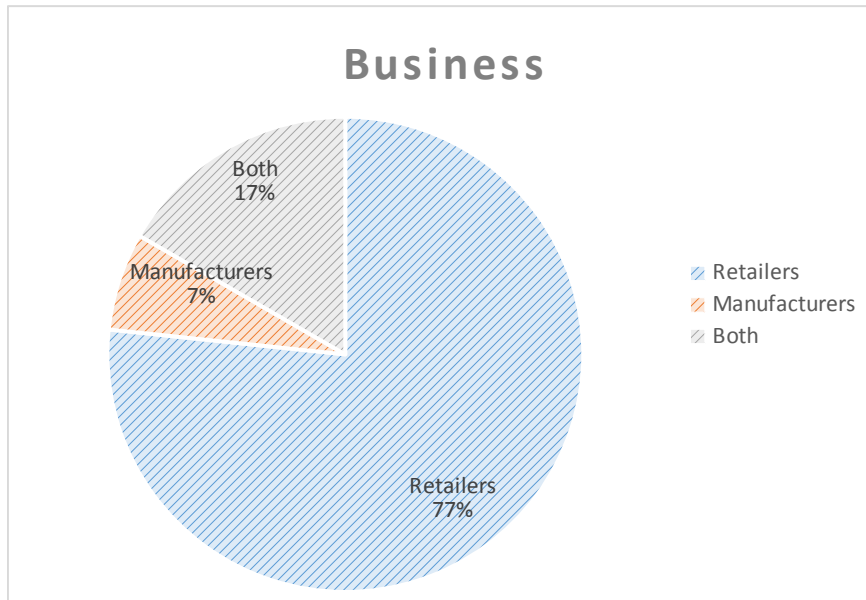
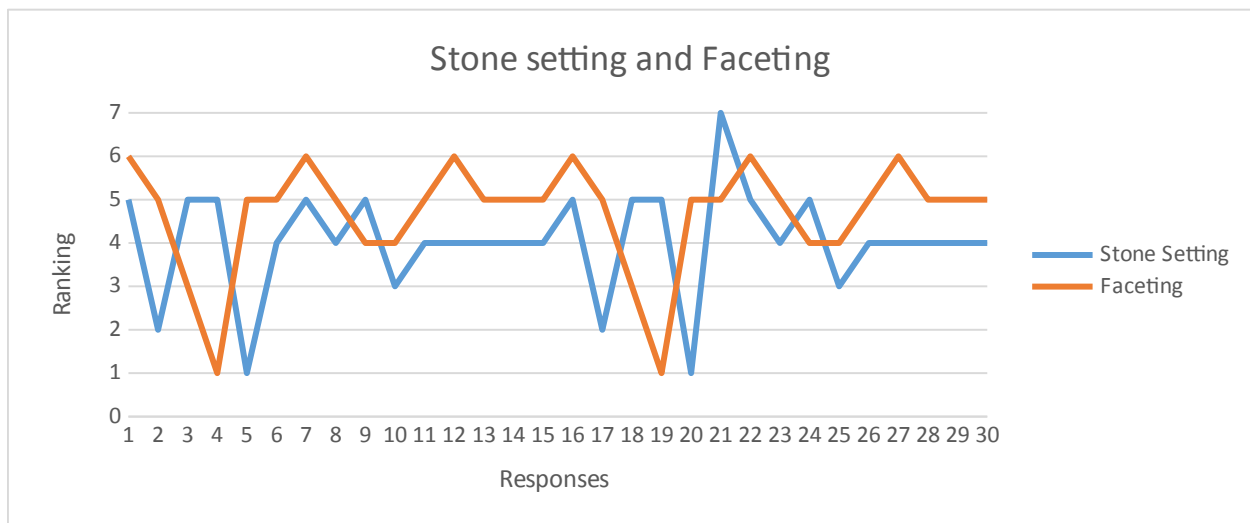
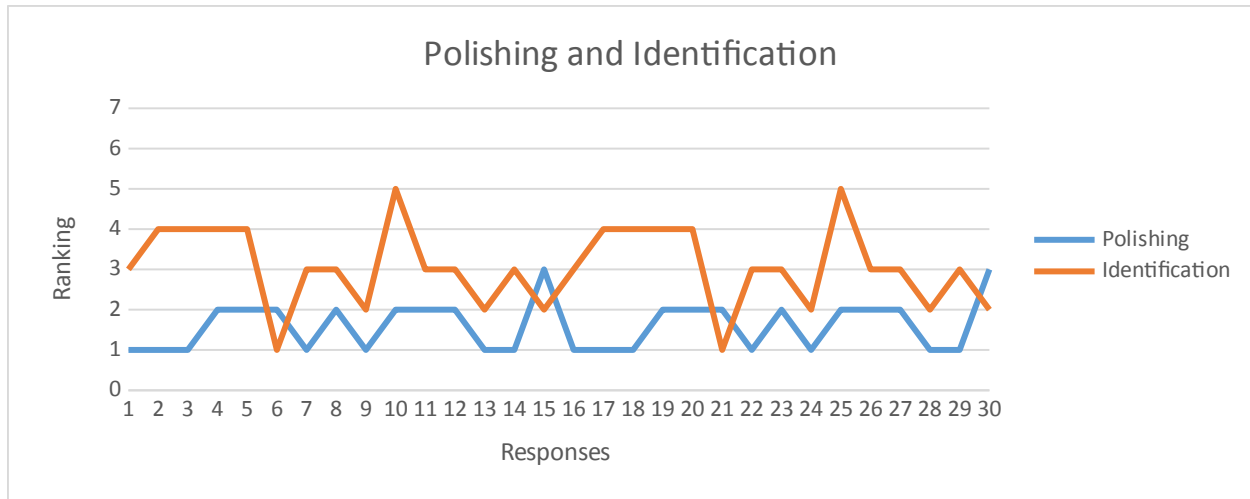


Exhibit 8: All these figures are based on our sample of 30 jewelers out of our population of 130 surveys.

No. of Participants	Items	Business	Demand	System
1	Gold Silver Diamond	Retailing	An experienced worker	Family /ustaad
2	Gold Silver Diamond	Retailing	An experienced worker	Family /ustaad
3	Gold and Silver	Retailing	An experienced worker	Family /ustaad
4	Gold and Diamond	Retailing	An experienced worker	Family /ustaad
5	Gems	Retailing	An experienced worker	Family /ustaad
6	Gold	Retailing	An experienced worker	Family /ustaad
7	Diamonds	Retailing	An experienced worker	Family /ustaad
8	Gold Silver Diamond	Retailing	An experienced worker	Family /ustaad
9	Silver and Artificial	Retailing	An experienced worker	Family /ustaad
10	Artificial	Retailing	An experienced worker	Family /ustaad
11	Gold and Silver	Retailing	An experienced worker	Family /ustaad
12	Gold ,Silver, Artificial	Both	An experienced worker	Family /ustaad
13	Gold ,Silver, Artificial	Manufacturing	An experienced worker	Family /ustaad
14	Gold ,Silver, Artificial	Both	An experienced worker	Family /ustaad
15	Gold and Artificial	Both	An experienced worker	Family /ustaad
16	Gold	Retailing	An experienced worker	Family /ustaad
17	All	Retailing	An experienced worker	Self learning
18	All	Retailing	An experienced worker	Family /ustaad
19	Gold	Retailing	An experienced worker	Family /ustaad

20	Gold and Diamonds	Retailing	An experienced worker	Family /ustaad
21	Gold and Silver	Retailing	An experienced worker	Family /ustaad
22	Gold	Both	An experienced worker	Self learning
23	Silver	Retailing	An experienced worker	Family /ustaad
24	Artificial	Retailing	An experienced worker	Self learning
25	Gold and Silver	Retailing	An experienced worker	Self learning
26	Gold Silver Diamond	Both	An experienced worker	Self learning
27	Silver	Retailing	An experienced worker	Self learning
28	Artificial	Manufacturing	An experienced worker	Self learning
29	All	Retailing	An experienced worker	Family /ustaad
30	Gold	Retailing	An experienced worker	Family /ustaad

Result of a Sample of 30 responses									
No. of Participants	Polishing	Identification	Stone Setting	Faceting	Carving	Manual Aided Design	Computer Aided Design		
1	1	3	5	6	2	4	7		
2	1	4	2	5	6	3	7		
3	1	4	5	3	7	2	6		
4	2	4	5	1	6	3	7		
5	2	4	1	5	6	3	7		
6	2	1	4	5	6	3	7		
7	1	3	5	6	7	2	4		
8	2	3	4	5	6	1	7		
9	1	2	5	4	6	3	7		
10	2	5	3	4	6	1	7		
11	2	3	4	5	6	1	7		
12	2	3	4	6	7	1	5		
13	1	2	4	5	6	3	7		
14	1	3	4	5	6	2	7		
15	3	2	4	5	6	1	7		
16	1	3	5	6	2	4	7		
17	1	4	2	5	6	3	7		
18	1	4	5	3	7	2	6		
19	2	4	5	1	6	3	7		
20	2	4	1	5	6	3	7		
21	2	1	7	5	6	4	3		
22	1	3	5	6	7	2	4		
23	2	3	4	5	6	1	7		
24	1	2	5	4	6	3	7		
25	2	5	3	4	6	1	7		
26	2	3	4	5	6	1	7		
27	2	3	4	6	7	1	5		
28	1	2	4	5	6	3	7		
29	1	3	4	5	6	2	7		
30	3	2	4	5	6	1	7		
	48	92	121	140	178	67	194		



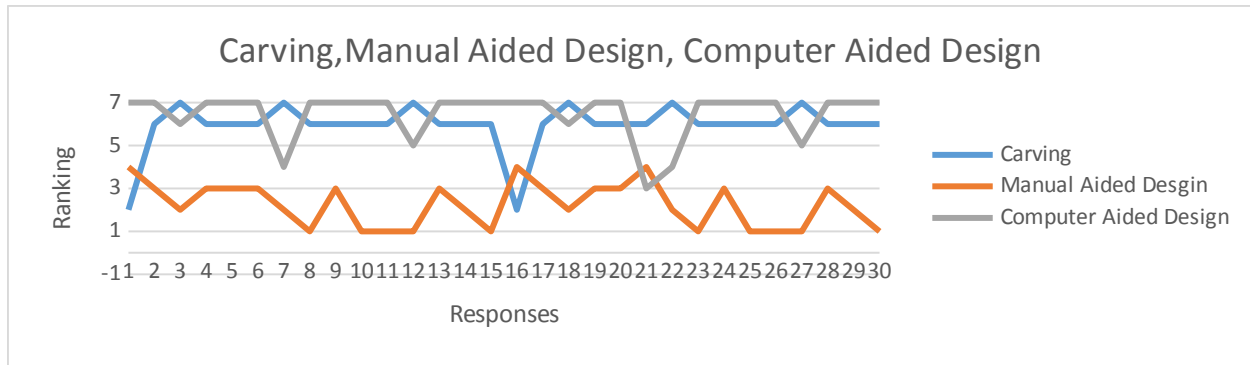
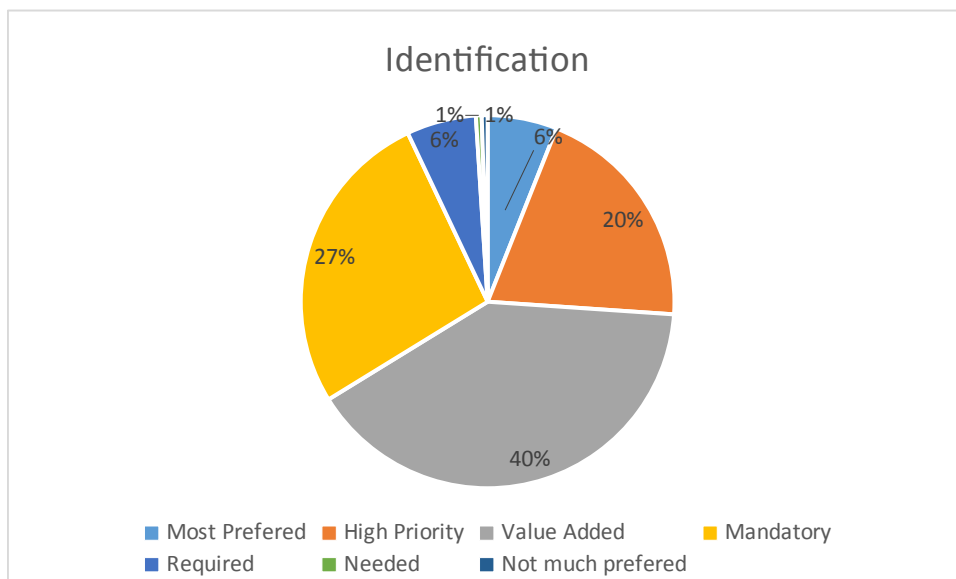


Exhibit 9:



Identification	Percentage
Most Preferred	6%
High Priority	20%
Value Added	40%
Mandatory	27%
Required	6%
Needed	0.5%
Not much preferred	0.5%
	100%

Exhibit 10:

Computer Aided Design	Percentage
Most Preferred	1%
High Priority	1%
Value Added	3%
Mandatory	7%
Required	7%
Needed	7%
Not much preferred	75%
	100%

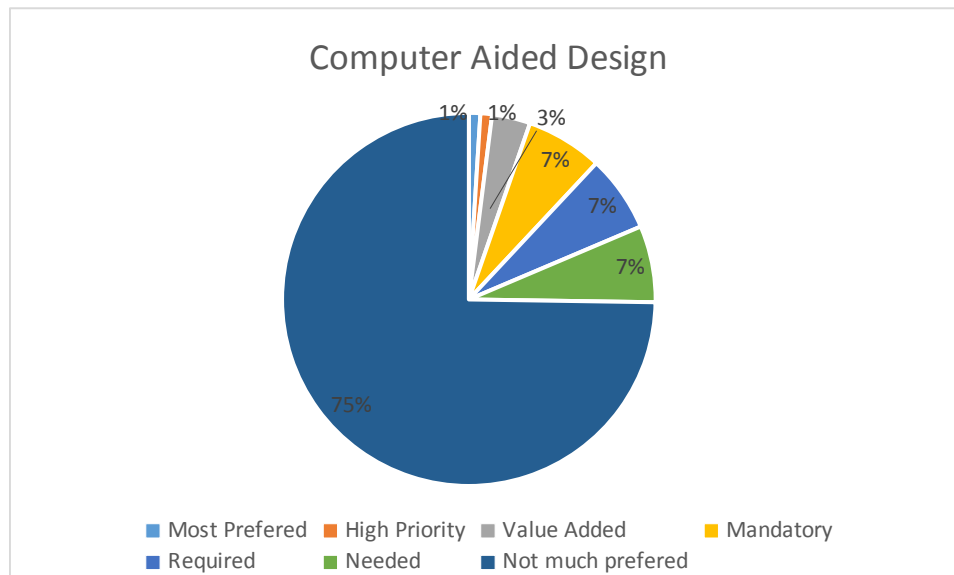


Exhibit 11:

Manual Aided Design	Percentage
Most Preferred	33%
High Priority	20%
Value Added	35%
Mandatory	10%
Required	0.5%
Needed	0.5%
Not much preferred	0.5%
	100%

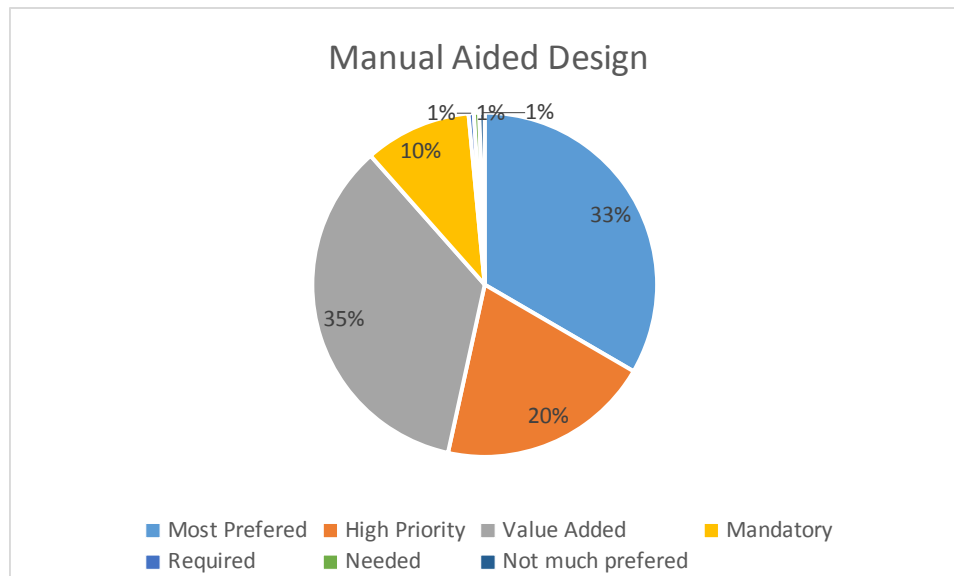


Exhibit 12:

Stone setting	Percentage
Most Preferred	7%
High Priority	7%
Value Added	7%
Mandatory	44%
Required	34%
Needed	1%
Not much preferred	1%
	100%

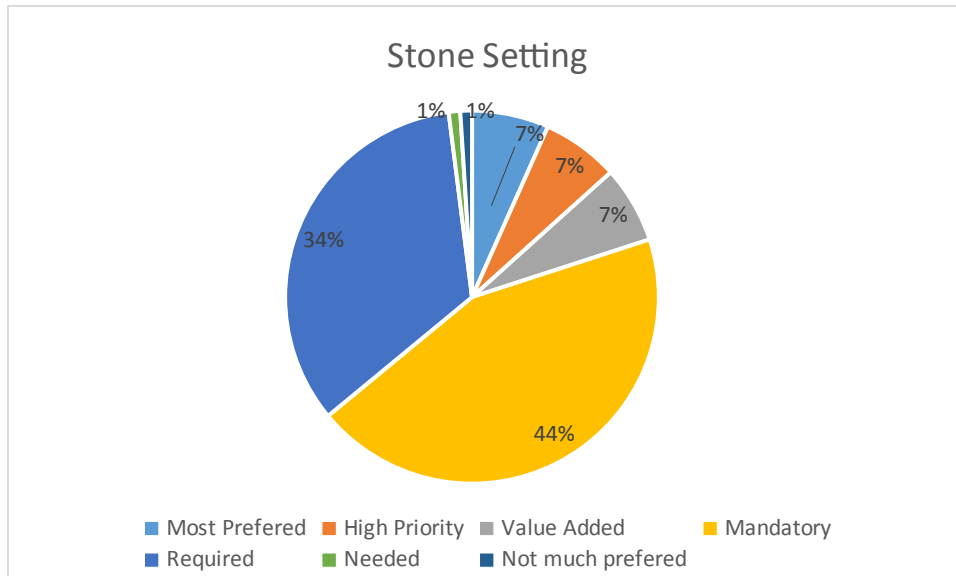
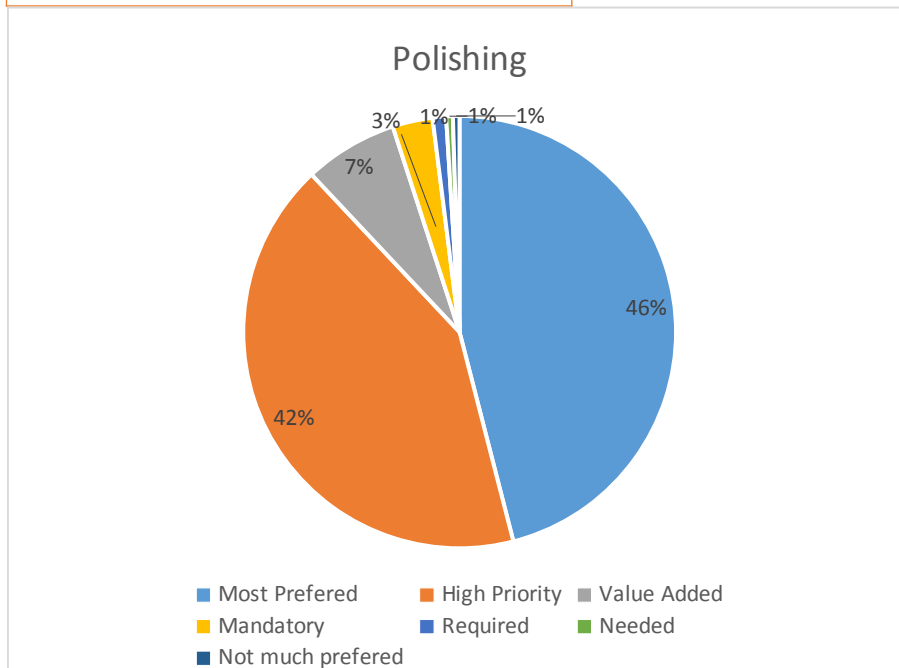
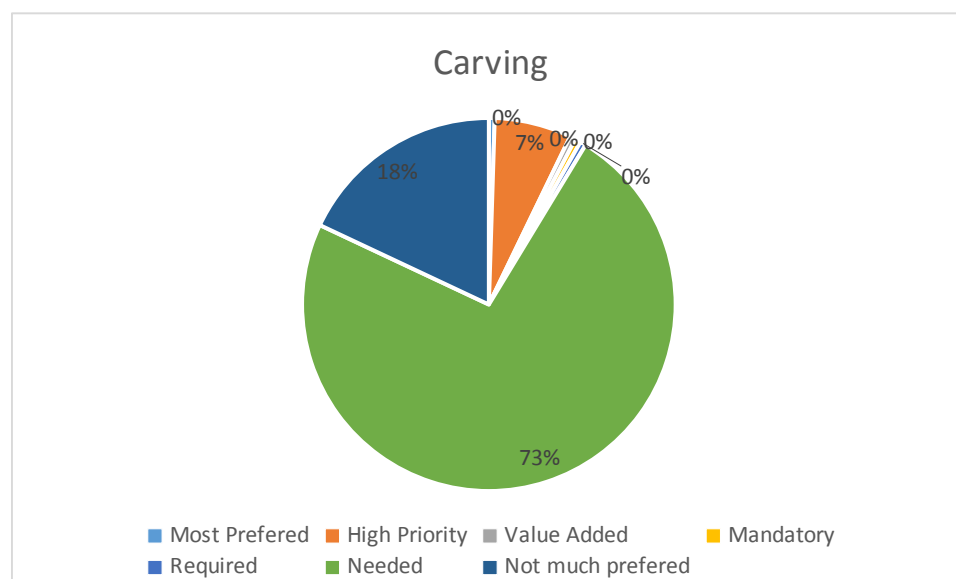


Exhibit 13:

Polishing	Percentage
Most Preferred	46%
High Priority	42%
Value Added	7%
Mandatory	3%
Required	1%
Needed	0.5%
Not much preferred	0.5%
	100%



Carving	Percentage
Most Preferred	0.5%
High Priority	7%
Value Added	0.5%
Mandatory	0.5%
Required	0.5%
Needed	73%
Not much preferred	18%
	100%



Faceting	Percentage
Most Preferred	6%
High Priority	0.5%
Value Added	7%
Mandatory	13%
Required	53%
Needed	20%
Not much preferred	0.5%
	100%

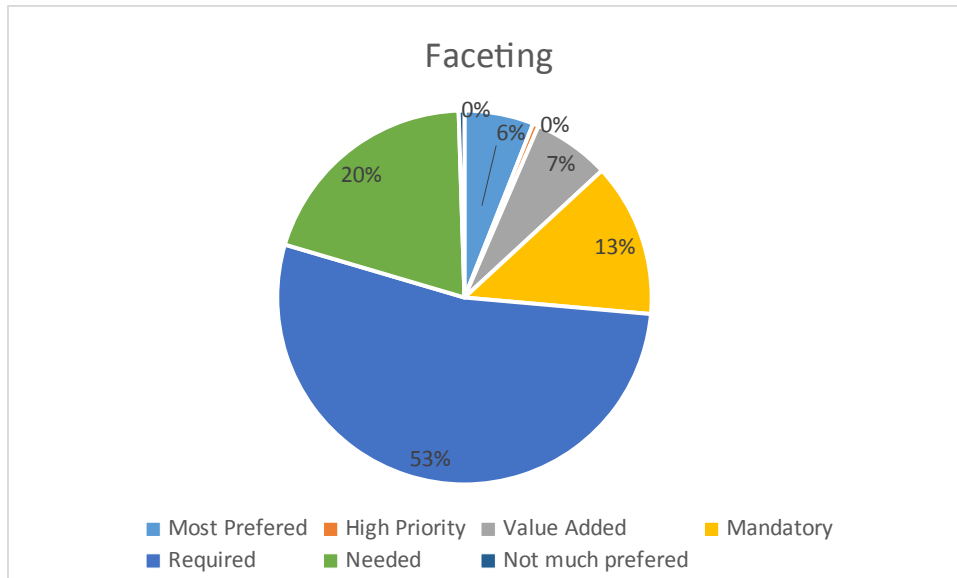


Exhibit 14: *Wax Model Making*



Exhibit 15: MPX Photo Impact Printers



Exhibit 16: MDX-40A CNC Mill/CNC Machine



Glossary

Azurite: a soft, deep blue copper mineral produced by weathering of copper ore deposits.

Agate: a cryptocrystalline variety of silica, chiefly chalcedony, characterized by its fineness of grain and brightness of color.

Calcite Dichroscope: an important pocket instrument used in the field of geology, and can be used to test transparent gemstones (crystals).

Casting Centrifuge: a casting technique that is typically used to cast thin-walled cylinders.

Crystallography: the experimental science of determining the arrangement of atoms in the crystalline solids (see crystal structure).

Crystal Structure: a unique arrangement of atoms, ions or molecules in a crystalline liquid or solid.

Diopside: a monoclinic pyroxene mineral with composition $\text{MgCaSi}_2\text{O}_6$.

Dremel: an American brand of power tools known primarily for its rotary tools.

Emery Paper: a type of sandpaper that can be used for sanding down hard and rough surfaces.

Faceting: the process of removing parts of a polygon, polyhedron or polytope, without creating any new vertices.

Garnet: a group of several closely related minerals.

Geiger Counters: devices that detect radioactivity.

Gemmology: the science dealing with natural and artificial gems and gemstones.

Internal Rate of Return: The discount rate often used in capital budgeting that makes the net present value of all cash flows from a particular project equal to zero.

Lapping: a machining process, in which two surfaces are rubbed together with an abrasive between them, by hand movement or using a machine.

Mohs Hardness: a qualitative ordinal scale which characterizes the scratch resistance of various minerals through the ability of a harder material to scratch a softer material.

Paragmite: a complex inosilicate mineral of the amphibole group.

Peridot: a well-known and ancient gemstone, with jewelry pieces dating all the way back to the Pharaohs in Egypt.

Polariscope: an instrument for measuring or exhibiting the polarization of light or for examining substances in polarized light.

Pleochroism: an optical phenomenon in which a substance appears to be different colors when observed at different angles, especially with polarized light.

Quartz: the second most abundant mineral in the Earth's continental crust.

Rose Quartz: a rose pink colored quartz (see quartz).

Vibrasonic Tumbler: a fast and efficient tumbler.

Vinyl: synthetic resin or plastic consisting of polyvinyl chloride or a related polymer used for wallpapers and other covering materials and for gramophone records.

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